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The Strategic Case for Oil and Gas to 2050+

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Long-Run Projections For Global Oil Demand Typically Miss This Crucial Fact

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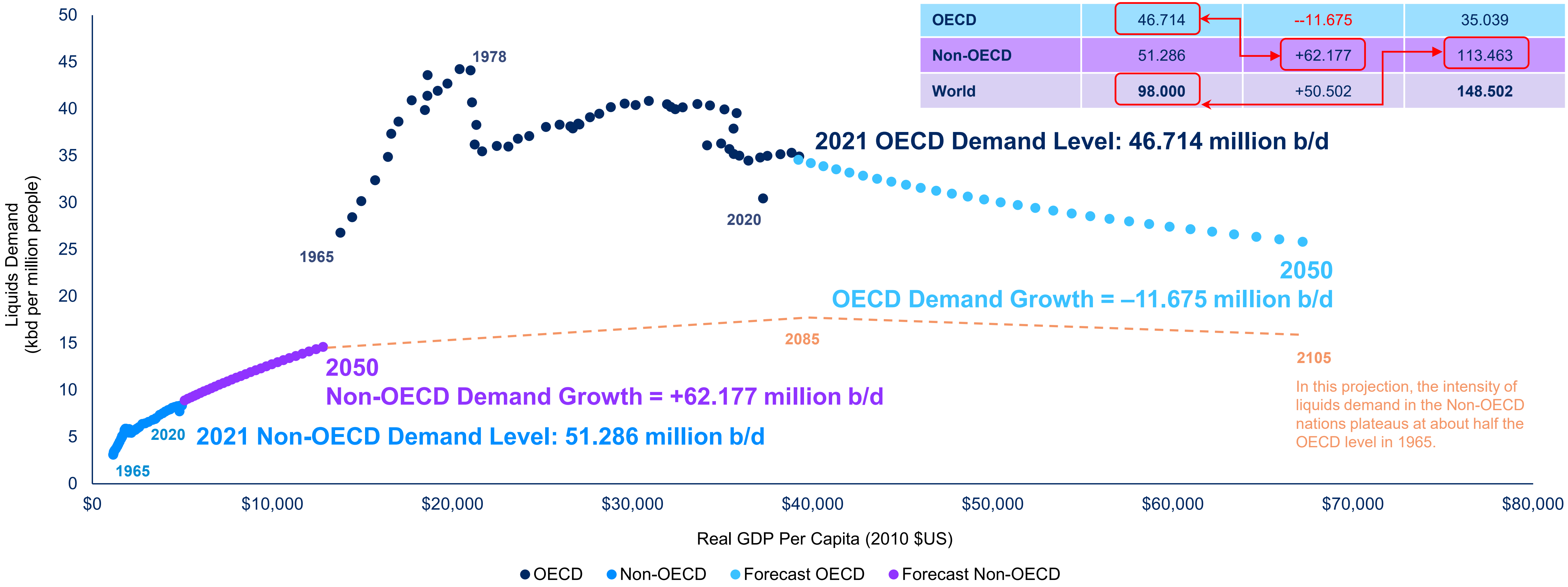
Even if all oil demand in all rich countries disappeared by 2050, global oil demand would most likely still be bigger than today

Liquids Demand	2021 Demand (million b/d)	Projected Change (Baseline)	2050 Demand (million b/d)
OECD	46.714	--11.675	35.039
Non-OECD	51.286	+62.177	113.463
World	**98.000**	+50.502	**148.502**

More People + More Wealth = More Liquids Demand (Even With Meaningful Conservation)



Source: *IEA World Energy Outlook 2020* (October 10, 2020), *IEA Net Zero by 2050: A Roadmap for the Global Energy Sector* (May 17, 2021), *BP Energy Outlook 2020* (September 14, 2020), *BNEF New Energy Outlook 2021* (July 2021), bp, World Bank, UN, BLR. Note: OECD = Organization for Economic Cooperation and Development (38 wealthy countries). Non-OECD = all countries and territories other than OECD countries (n=180 developing economies). Note: these are projections not forecasts. These projections are based on unconstrained trend growth rates in population and real GDP across 218 countries and territories. As such, they map a useful reference scenario for the "business as usual" baseline for global liquids demand, against which alternative scenario projections may be measured. Alternative scenarios would rely on changes in demographic, economic, technological, geopolitical, behavioral, policy, and other factors. One core purpose of this exercise is to demonstrate that extant trends in Non-OECD would lift its 2050 intensity of liquids demand to a level still far below the intensity of liquids demand in the OECD countries in 1965, or 57 years before the date of this presentation. Notice that the "business as usual" projections suggest global liquids demand will be 148.5 million b/d in 2050, a net increase of 50.5 million b/d (+52%) from 2021, as a 62 million b/d gain in Non-OECD offsets a 12 million b/d drop in OECD. In this case, if all current OECD demand (46.7 million b/d) vanished, global liquids demand would still grow to 113 million b/d. These boundary conditions illustrate why most credible long-range forecasts for 2050 fall between 130 million b/d and 150 million b/d.

The Demographic Momentum for Natural Gas Is Even Stronger

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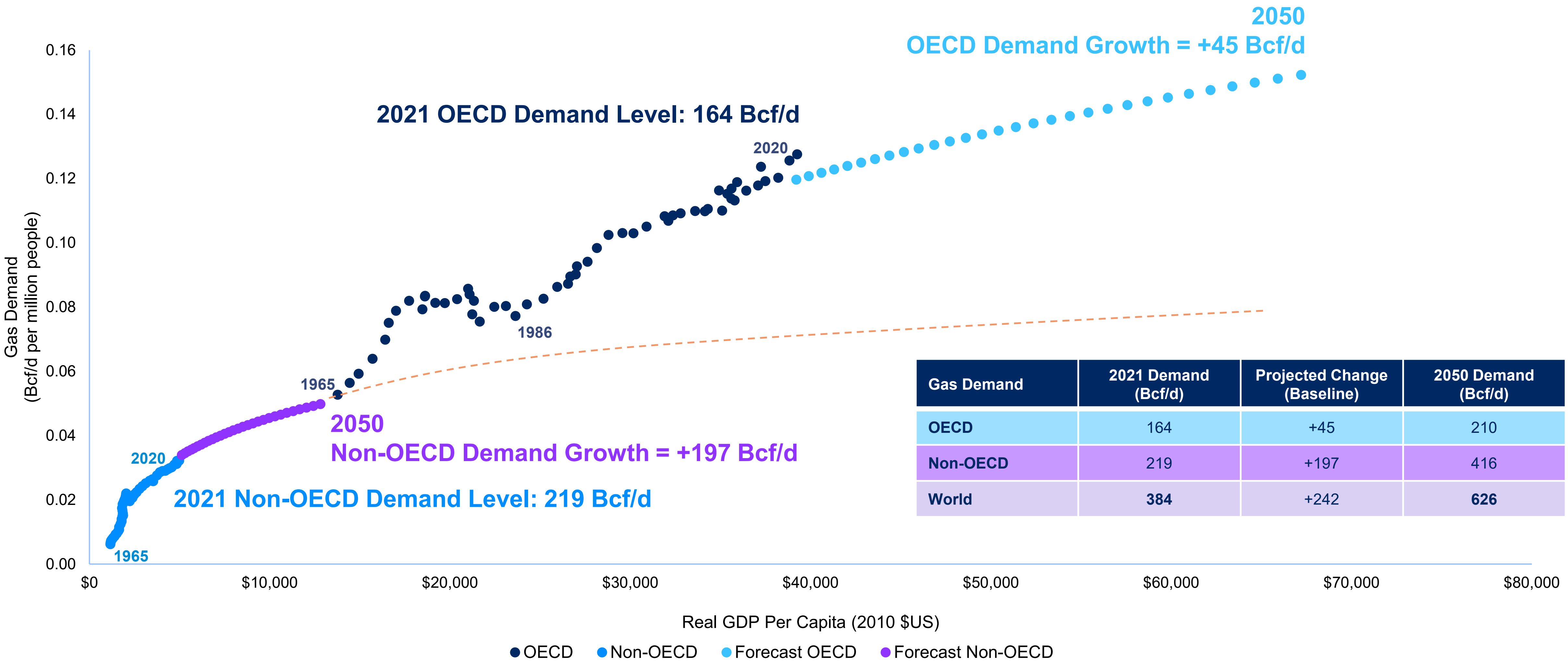

Gas Demand	2021 Demand (Bcf/d)	Projected Change (Baseline)	2050 Demand (Bcf/d)
OECD	164	+45	210
Non-OECD	219	+197	416
World	**384**	+242	**626**

Source: IEA, bp, World Bank, UN, BLR. Note: these are projections not forecasts. These projections are based on unconstrained trend growth in population and real GDP across 218 countries and territories. As such, they map a useful reference scenario for the “business as usual” baseline for global natural gas demand, against which alternative scenario projections may be measured. Alternative scenarios would rely on changes in demographic, economic, technological, geopolitical, behavioral, policy and other factors. One core purpose of this exercise is to demonstrate that extant trends in Non-OECD would lift its 2050 intensity of gas demand to a level roughly comparable with the intensity of gas demand in the OECD countries in 1965, or 57 years before the date of this presentation.

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Yet, Capital Is Starved Throughout The Global Oil & Gas Supply Chain…

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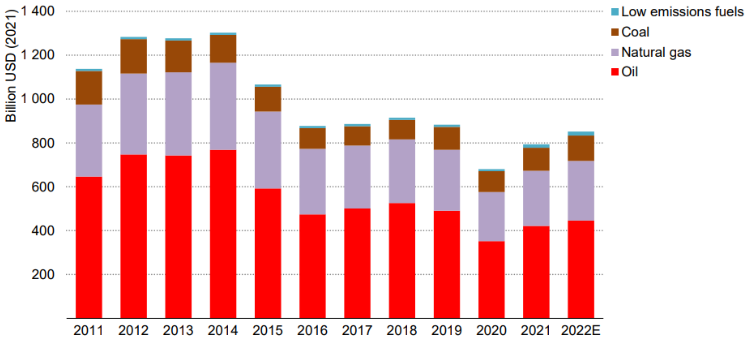
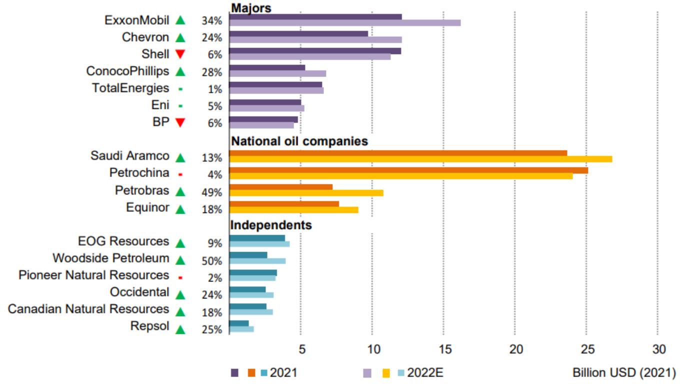
Global investment in fuels is set to rise in 2022, but remains below pre-pandemic levels

Global investment in fuel supply, 2011-2022E



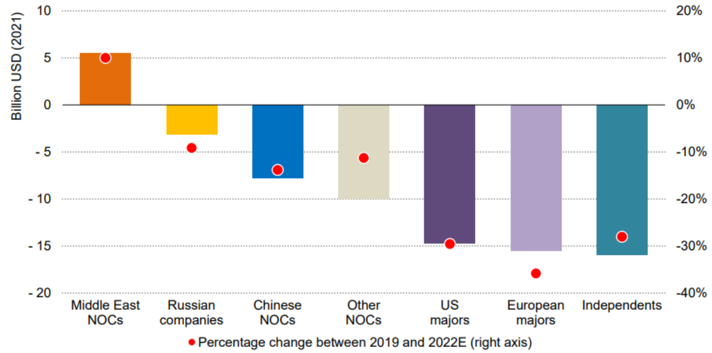
Total upstream oil and gas investment is set to rebound by around 10% in 2022, led by the US majors, independents and NOCs in the Middle East…

Upstream investment by selected oil and gas companies, 2021 and expected in 2022



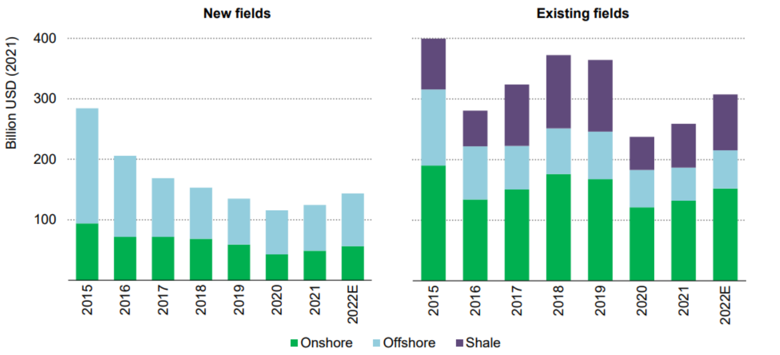
…but only Middle Eastern NOC spending has risen above pre-pandemic levels. Spending by majors and independents is still well below average levels seen in the last decade.

Estimated change in upstream spending by selected company types, 2019-2022E



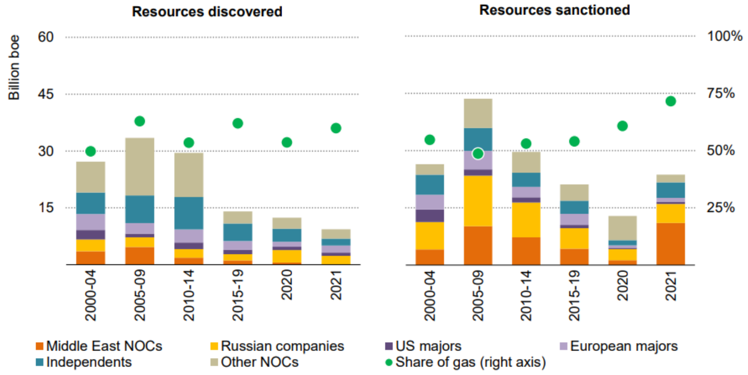
Investment in new fields is back on a rising trend but most upstream capital spending is still on existing fields and shale plays

Upstream investment in new and existing oil and natural gas fields



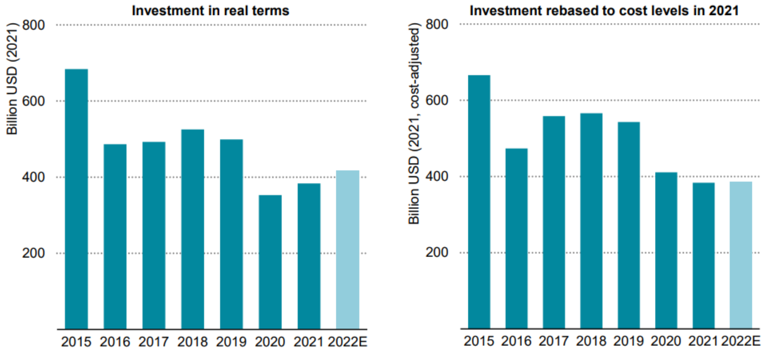
Conventional discoveries are falling but still significant, while decisions to sanction new developments picked up in 2021, with a focus on natural gas

Conventional oil and gas resources discovered and sanctioned, 2000-2021



The increase in upstream oil and gas spending in 2022 is largely a reflection of higher costs.

Global investment in upstream oil and gas in 2021, real terms (left) and rebased at constant 2021 costs (right), 2015-2022E



Source: *World Energy Investment 2022* (IEA).

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…Even As Oil Downstream Asset Retirements Mount Toward Record Highs

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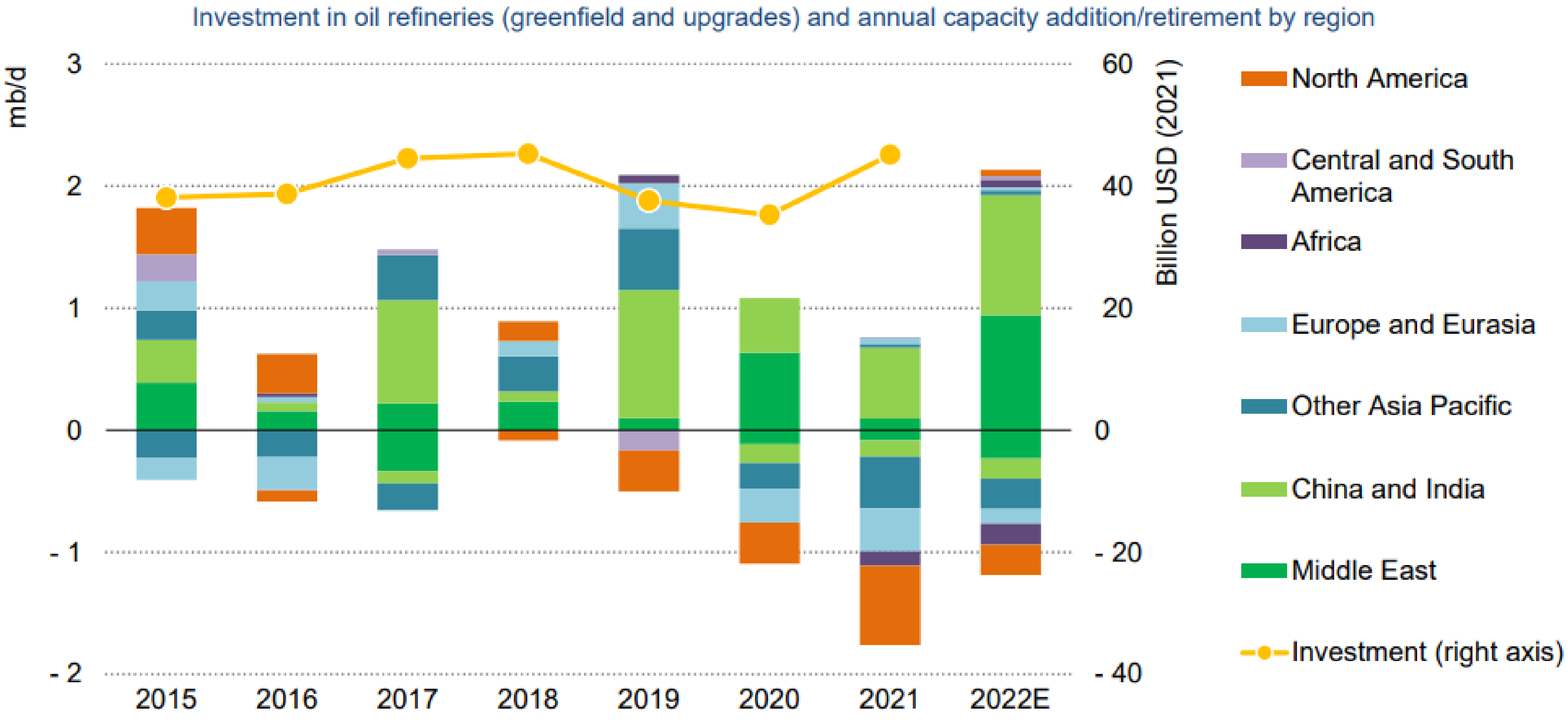
Investment in new refineries and upgrades increased by 30% in 2021, but a near-record level of refining capacity was retired



Source: *World Energy Investment 2022* (IEA).

OPEC Projections for Primary Energy Demand Growth, 2020-2045P

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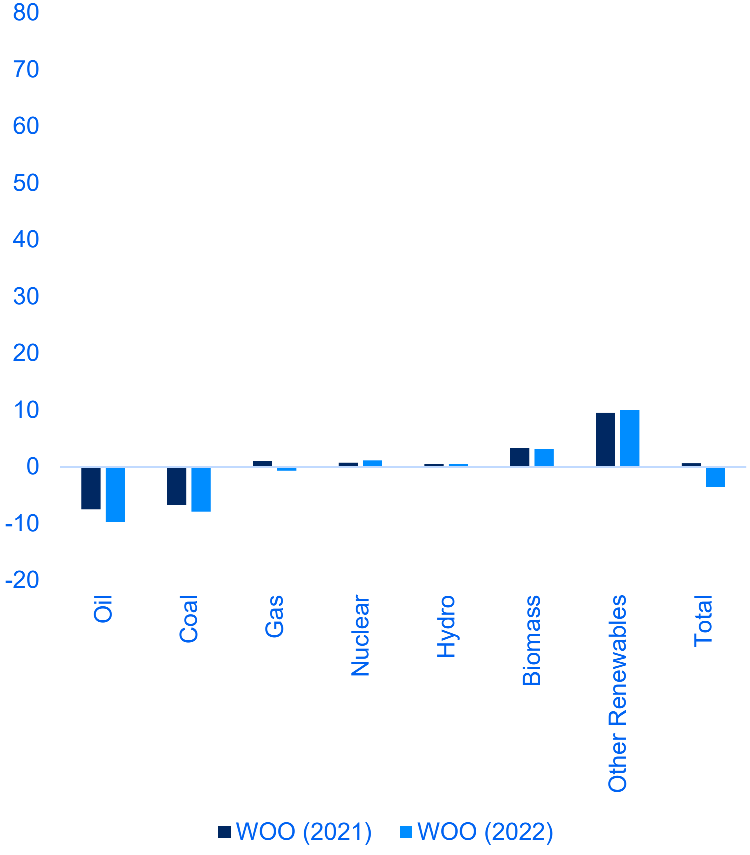
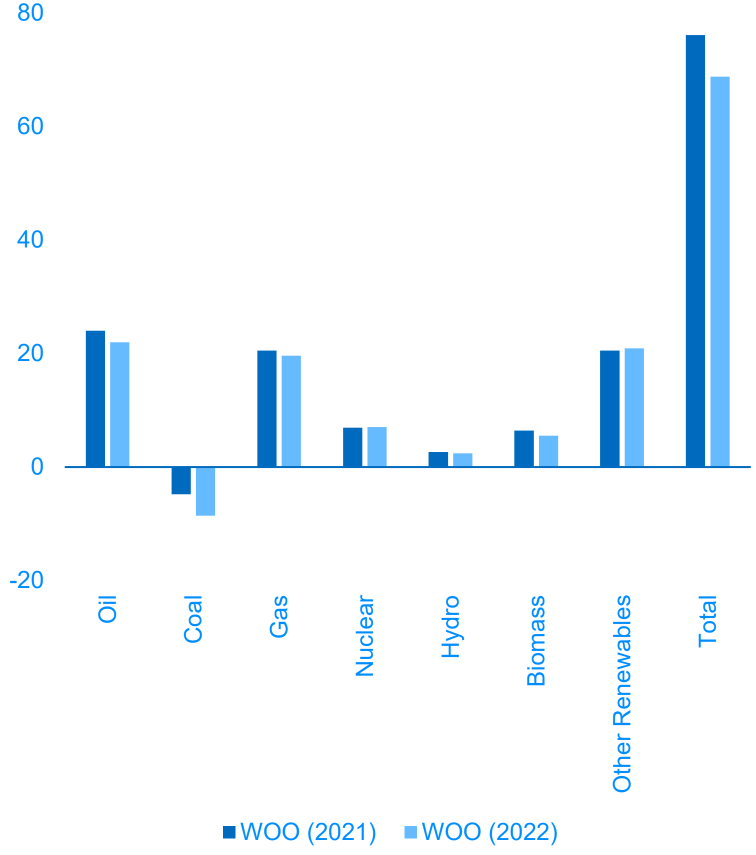
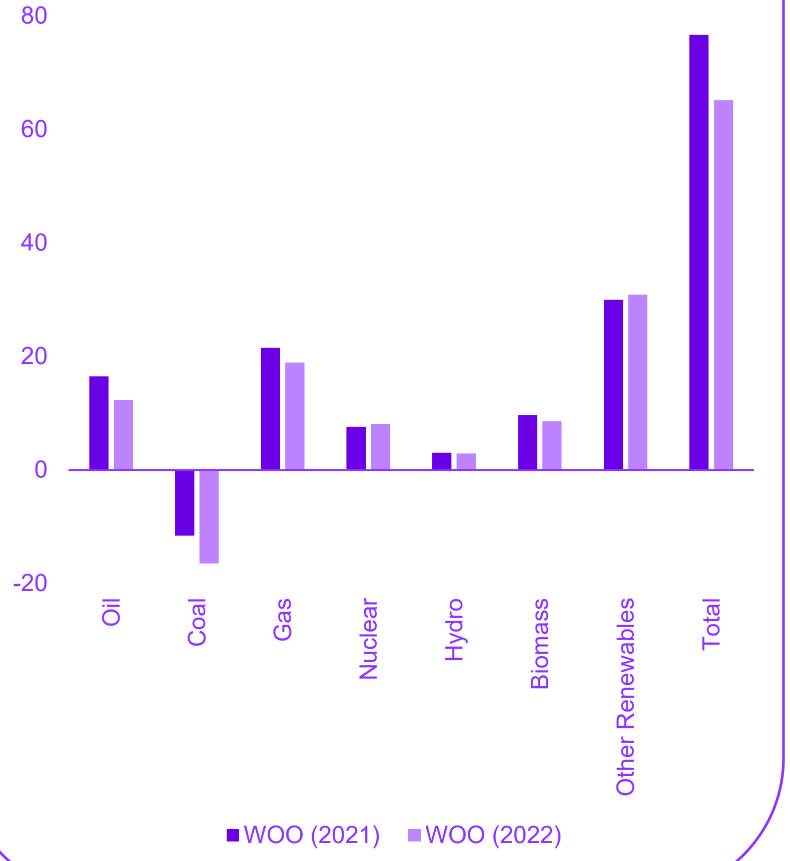
OECD
cumulative change in exajoule (EJ)



Non-OECD
cumulative change in exajoule (EJ)



World
cumulative change in exajoule (EJ)



Source: OPEC (*World Oil Outlook 2021* and *World Oil Outlook 2022*), Veriten. Note: OPEC published WOO 2022 on October 31, 2022.

IEA Projections for World Energy Supply Growth, 2020-2050P

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World Energy Supply Growth, 2020-2050P

cumulative change in exajoule (EJ) per year

IEA now says world energy supply needs to shrink by 6X as much as last year's estimate

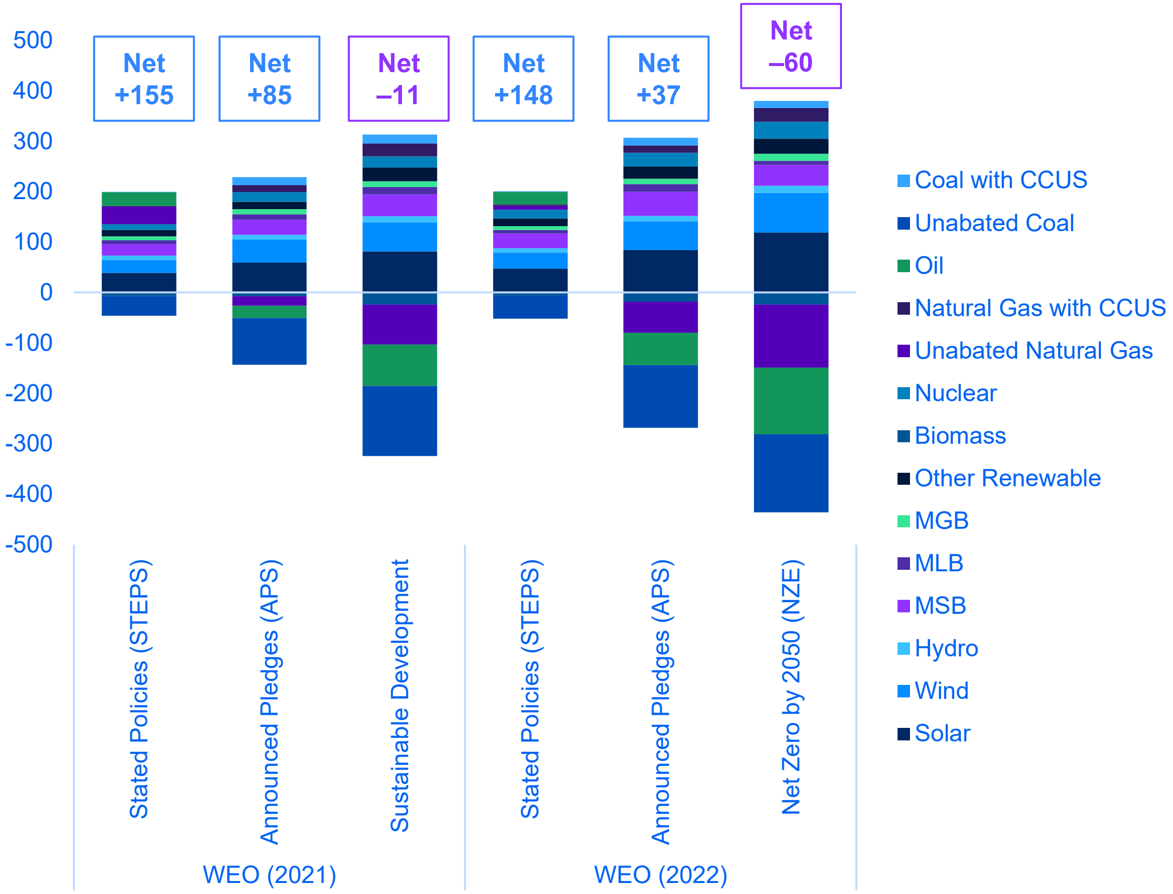

Table 5.1 ▷ Key energy indicators by scenario, 2010-2050

			STEPS		APS		NZE	
	2010	2021	2030	2050	2030	2050	2030	2050
Access (million people)								
Population without access to electricity	1 392	754	663	727	292	112	0	0
Population without access to clean cooking	2 916	2 386	1 880	1 601	783	535	0	0
Premature deaths from (million people):								
Ambient air pollution	n.a.	4.2	4.8	7.1	4.6	6.5	3.3	2.9
Household air pollution	n.a.	3.6	2.9	3.0	1.6	1.9	1.0	1.2
Energy-related CO_2 emissions (Gt)	32.9	36.6	36.2	32.0	31.5	12.4	22.8	0
CO_2 captured via CCUS	0	0.04	0.1	0.4	0.5	4.3	1.2	6.2
Primary energy supply (EJ)	542	624	673	740	636	629	561	532
Share of unabated fossil fuels	81%	79%	74%	61%	69%	34%	59%	10%
Energy intensity of GDP (GJ per USD 1 000, PPP)	5.1	4.3	3.4	2.2	3.2	1.9	2.9	1.6
Electricity generation (1 000 TWh)	22	28	35	50	36	61	38	73
CO_2 intensity of generation (g CO_2/kWh)	524	459	325	158	280	41	165	-5
Share of low-emissions generation	32%	38%	53%	74%	59%	91%	74%	100%
Total final consumption (EJ)	383	439	485	544	451	433	398	337
Share of unabated fossil fuels	69%	66%	64%	57%	61%	36%	56%	15%
Share of electricity in TFC	17%	20%	22%	28%	24%	39%	28%	52%

Notes: Gt = gigatonnes; CCUS = carbon capture, utilisation and storage; EJ = exajoule; GJ = gigajoule; PPP = purchasing power parity; TWh = terawatt-hour; kWh = kilowatt-hour; TFC = total final consumption. STEPS = Stated Policies Scenario; APS = Announced Pledges Scenario; NZE = Net Zero Emissions by 2050 Scenario.

Source: IEA (*World Energy Outlook 2021* and *World Energy Outlook 2022*), Veriten. Note: IEA published WEO-2022 on October 27, 2022. MGB = Moden Gaseous Bioenergy. MLB = Modern Liquid Bioenergy. MSB = Modern Solid Bioenergy. CCUS = Carbon Capture Utilization Storage. In energy content terms, +155 EJ is +25.3 billion barrels oil equivalent, or +69.4 million b/d oil equivalent, and –60 EJ is –9.81 billion barrels oil equivalent, or –26.9 million b/d oil equivalent.

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This Discrepancy Persists Due To Large Assumptions About "Peak Demand"

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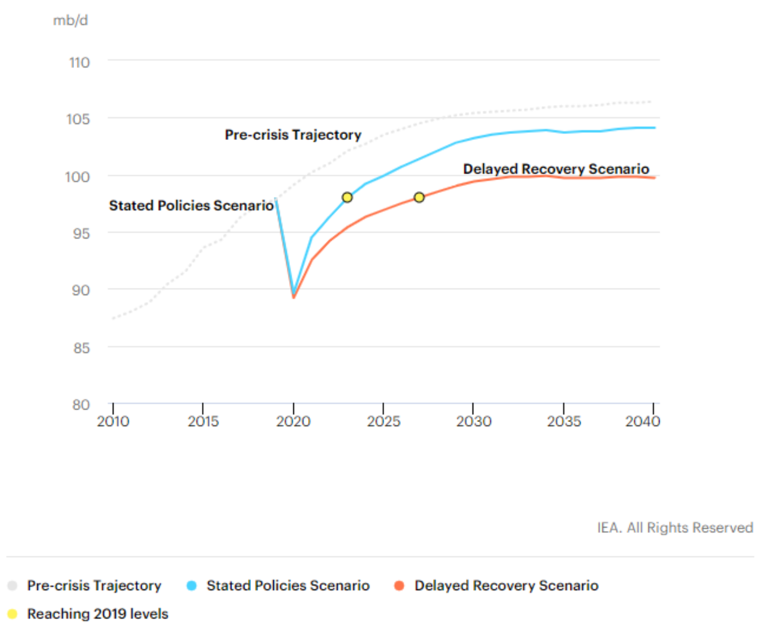
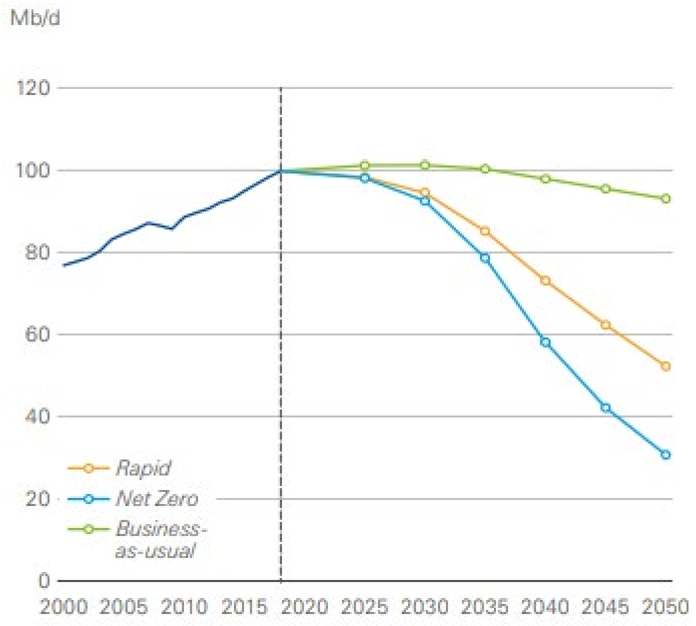
IEA (October 2020): A Long Plateau



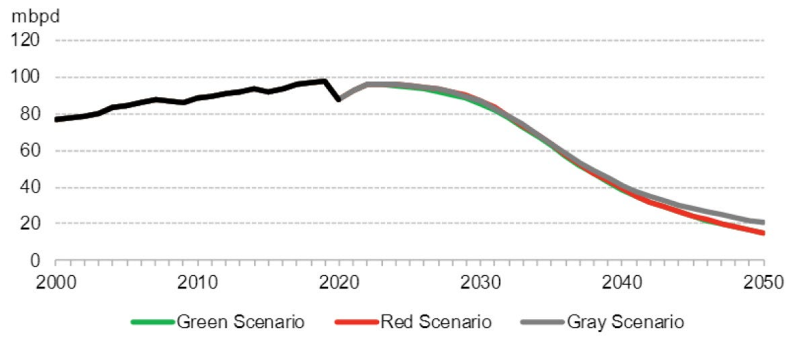
In May 2021, IEA sees 'no need' for investment in 'new oil supply' yet pegs demand at 100 mbd in 2040?

BP (September 2020): A Peak, Then Rapid Decline



BP asserts that global oil demand may have already peaked in 2019. Its Net Zero projection foresees 70% decline.

BNEF (July 2021): 80% Lost in All Scenarios



BNEF also claims US on-road vehicle fleet is 200 million units and will stay flat for 30 years.

Source: *IEA World Energy Outlook 2020* (October 10, 2020), *IEA Net Zero by 2050: A Roadmap for the Global Energy Sector* (May 17, 2021), *BP Energy Outlook 2020* (September 14, 2020), *Bloomberg New Energy Finance (BNEF) New Energy Outlook 2021* (July 2021), BLR.

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Real Events Are Rapidly Eroding Confidence In Those Large Assumptions

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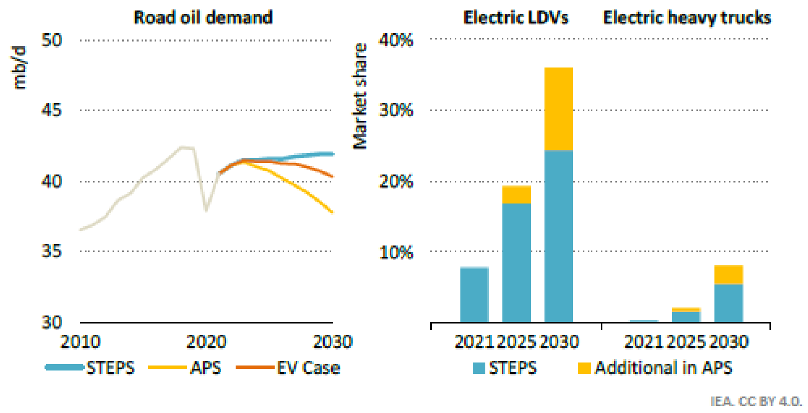
IEA (October 2022): Narrower Focus on Road Oil

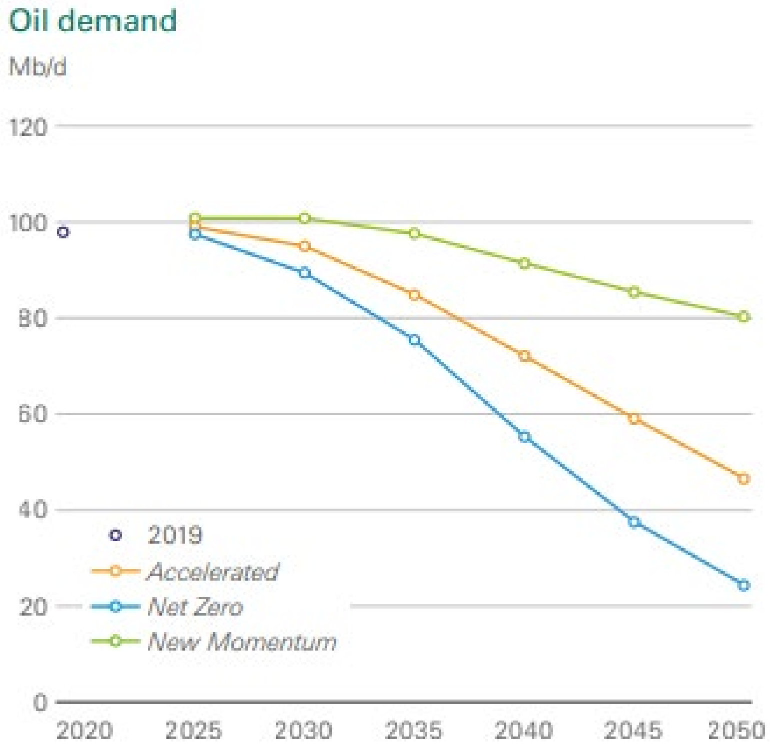
Figure 5.21 ▷ Global road transport oil demand by scenario, 2010-2030, and EV sales by scenario, 2021-2030



Oil demand in road transport increases slightly to 2030 in the STEPS despite the growth of EVs, with faster growth for electric LDVs and heavy trucks it would peak a decade earlier

Notes: EV = electric vehicle; LDVs = light-duty vehicles which include passenger cars and light trucks. Heavy trucks include medium- and heavy-freight vehicles. Electric refers to battery electric vehicles as well as plug-in hybrid vehicles. The EV case shows the STEPS trajectory with the penetration of EVs at similar levels of the APS.

BP (June 2022): Sees Even Steeper Declines



BNEF (July 2022): Sharp Shift In Tone

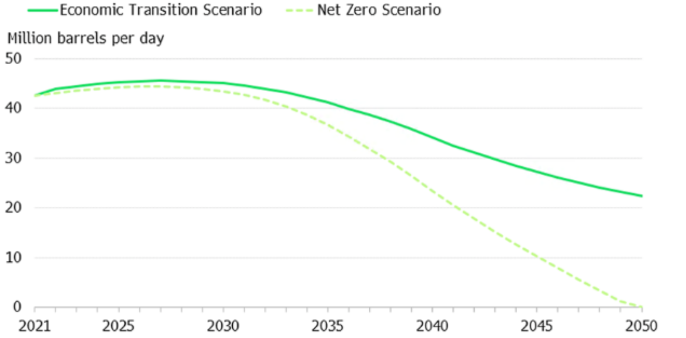
5. Nowhere near zero

While the long-term decline of gasoline and diesel is rapidly approaching, the world is nowhere near on track to eliminate fossil fuel use in the road transport sector.

Even with close to 2.4 billion zero-emission vehicles driving around in 2050, demand for road fuels is set to remain above 20 million barrels per day under BNEF's Economic Transition Scenario.

Down But Not Out

Economics alone will not be enough to eliminate fossil fuels from road transport



Source: BloombergNEF, International Energy Agency. Note: Road fuels include fossil- and bio-derived gasoline, diesel and liquefied petroleum gas. Scenarios above include an estimate for buses.

Source: *IEA World Energy Outlook 2022* (October 2022), BP (2022 Energy Outlook), BNEF (July 5, 2022)

Top Ten Biases that Propagate Error in Projections for Oil Product Demand

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1. **Incompleteness Bias:** undercount current U.S. on-road vehicle stock (e.g., 276M in '19 not 200M)
2. **'Where' Bias:** ignore off-road demand, including all construction and farm equipment; also applies to light duty vehicles vs heavy trucks
3. **Use Case Bias:** ignore all non-fuel uses (NGLs and petrochemicals), 15% - 25% of some markets
4. **Fixed Pie Bias:** assume static market size and 1-for-1 substitution of fuel source, rather than seeing growth of total pies (population, vehicles, VMT)
5. **'Not in Same World' Bias:** projections that may or may not have independent merit, but are incompatible with each other
6. **Own Worldview Bias:** project urban experience into regions that are and will remain suburban or rural; assume same average age for other states
7. **'Like Them' Bias:** assume disruption in natural resources must produce extinction outcomes comparable to what happens in the tech sector
8. **Time Bias:** projection of desired timetable (fast) on demand-side processes that realistically must take longer on cost (capital stock replacement cycles)
9. **Confirmation Bias:** extrapolate true facts into distorted/untrue portraits of current & future trends
10. **Sample Bias:** assume rates of uptake in Oregon and California will be matched in Georgia

Source: BLR, Veriten.

Fast Uptake of Battery Electric Vehicles (BEV) Is Not Fast Enough by 2050

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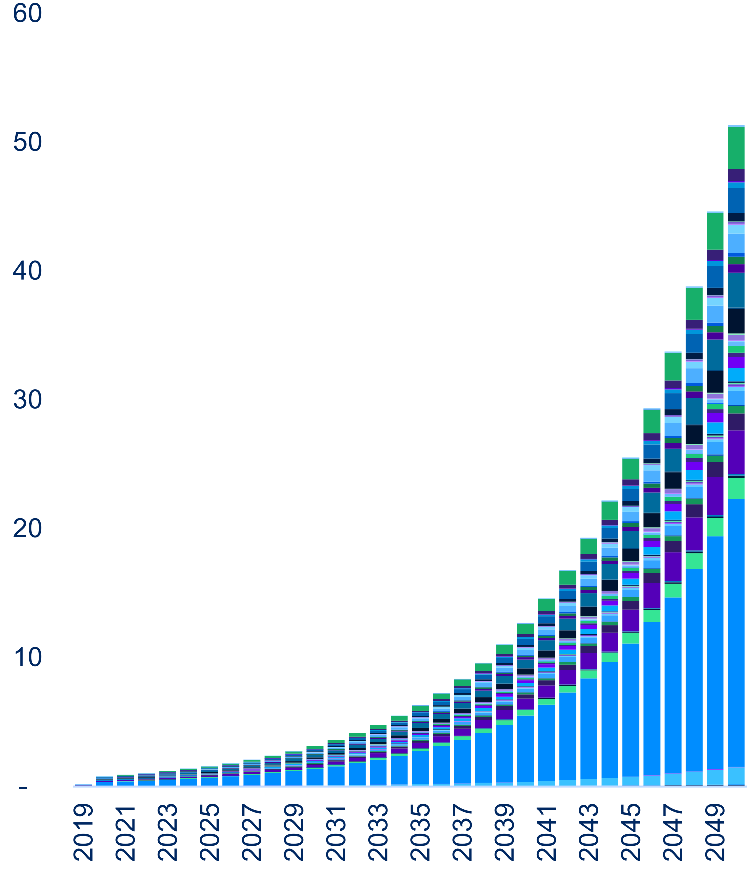
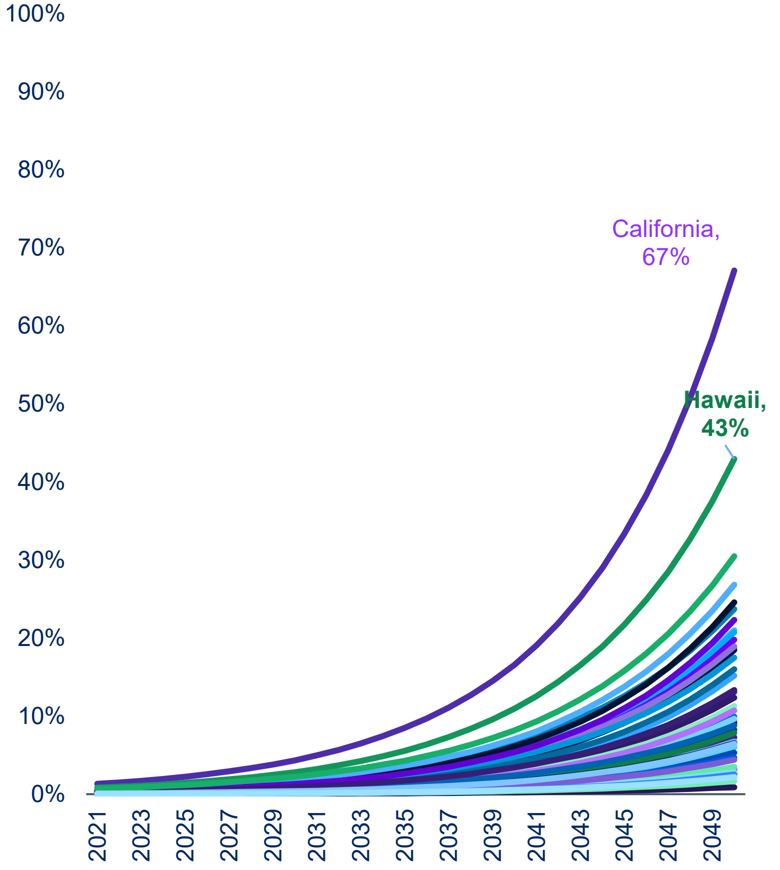
U.S. BEV Uptake, 15% CAGRs
million units in the U.S. on-road vehicle stock



U.S. BEV Penetration, 15% CAGRs
share of on-road total vehicle stock by state



Georgia 31-Oct-2022	Count and % of Total Registrations
Total Registrations	10,489,214
Passenger Vehicles	6,678,889 (63.7%)
Trucks	2,201,370 (21.0%)
Trailers	1,352,911 (12.9%)
Motorcycles	218,024 (2.1%)
Buses	37,990 (0.4%)
Other	30 (0.0%)
Electric Vehicles	**47,059 (0.4%)**

Source: FHWA/DOT, IHS-Markit, BLR. Note: in 2021, the average life of U.S. on-road vehicles reached 12 years, a new high. For households with incomes below the national median, vehicle life is averaging closer to 18 years. The BEV penetration rate in the United States was 0.37% as of June 2021, ranging among states from a high at 1.33% in California to a low of 0.02% in North Dakota. Required 2021-2050 CAGRs to achieve 100% BEV penetration in the total projected on-road vehicle stock by region are: Midcontinent (26%), Southeast (25%), Middle (28%), New England (25%), Atlantic (22%), and West (19%). **In Georgia, five counties account for two-thirds of the state's EV fleet: Fulton, Gwinnett, Cobb, DeKalb, and Chatham. There are 159 counties in Georgia.**

Best Models Readily Find Falling Demand Scenarios, But Not Zero Demand

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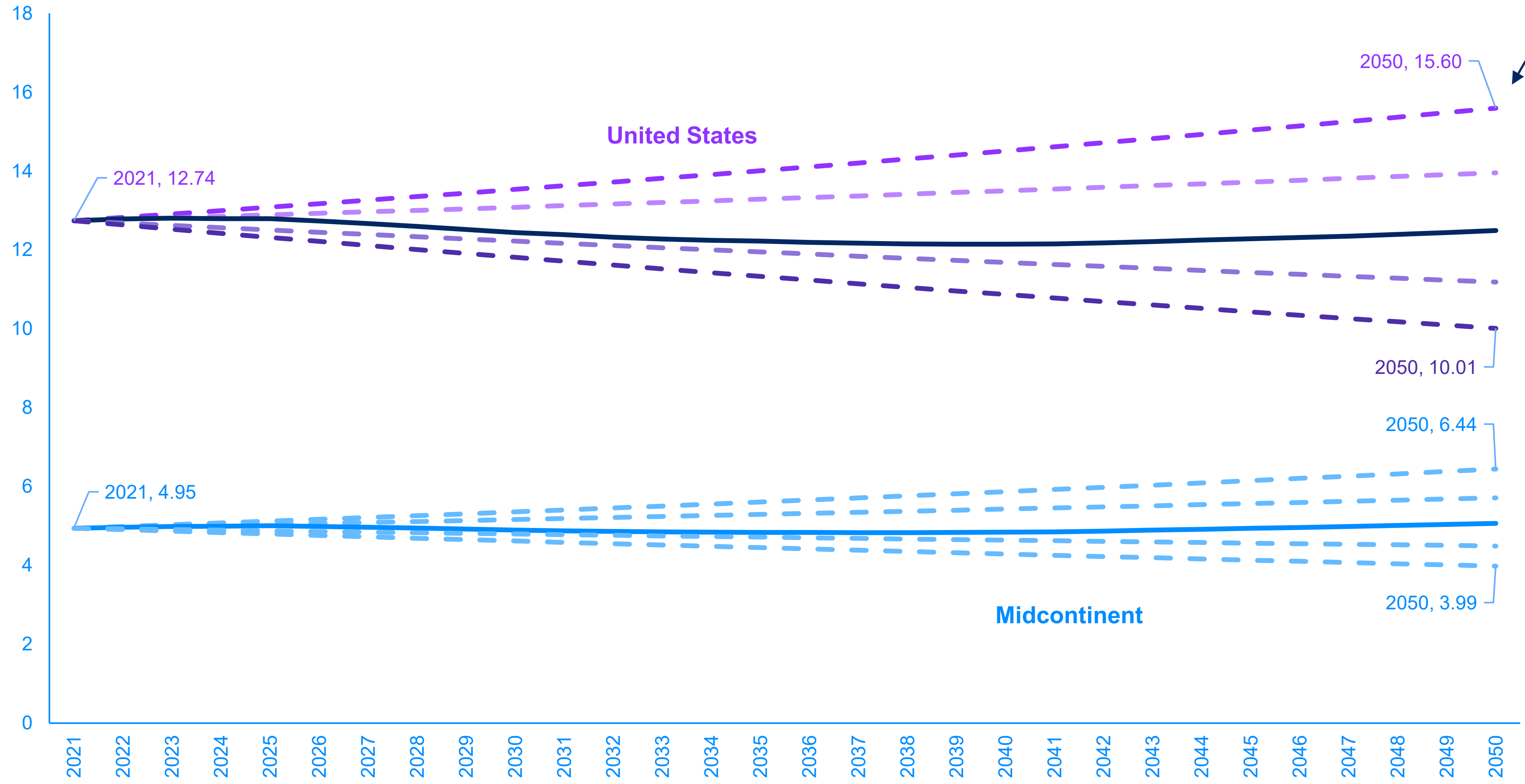
U.S. Gasoline Plus Distillate Demand Paths by Scenario
million b/d



95% confidence intervals also see upside scenarios, even in projections with non-normal probability distributions

Source: EIA, BLR.

Price Instability Forces Higher Policy Rates and Balance Sheet Contraction

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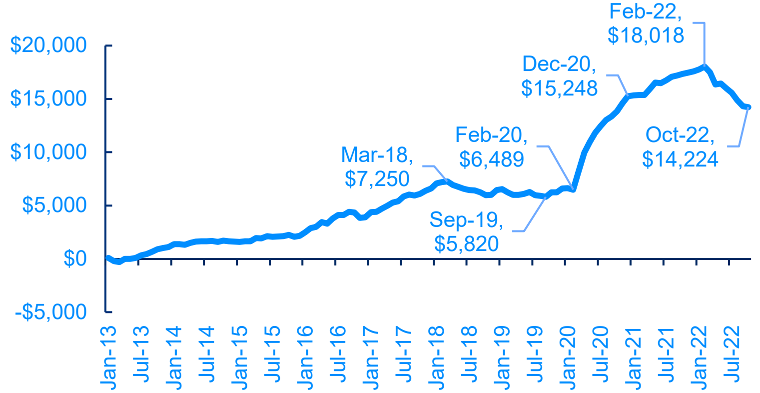
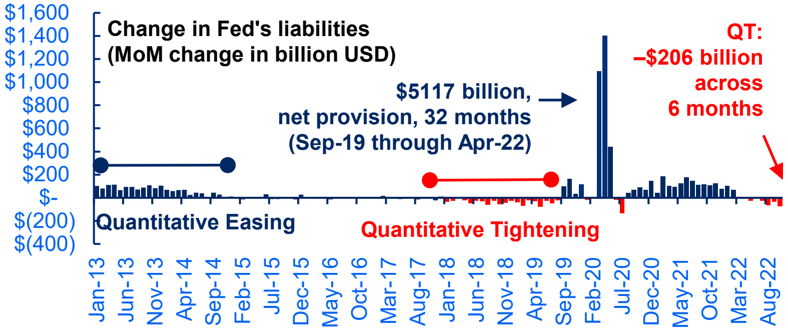
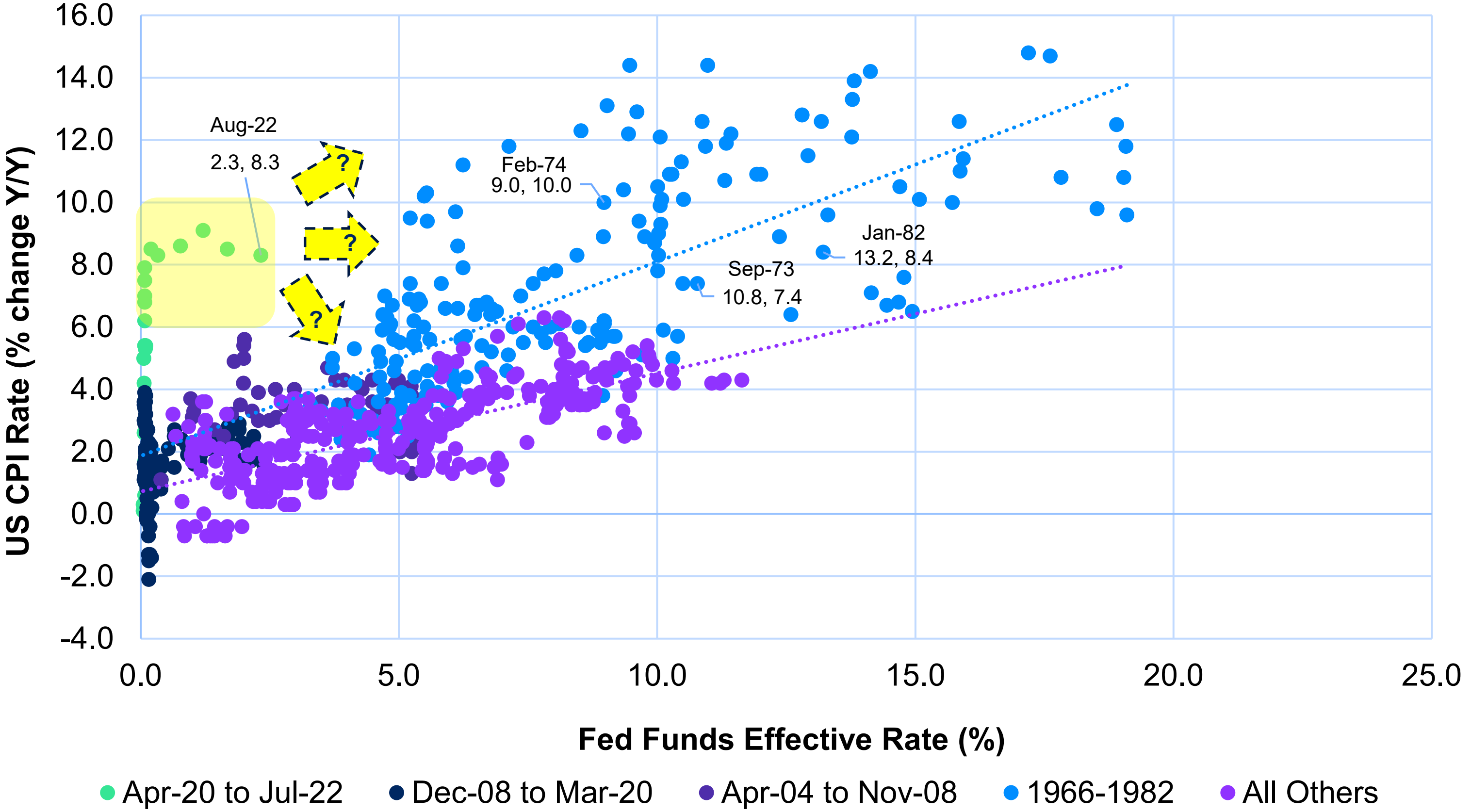
Growth in Four Central Banks' Balance Sheets
billion USD cumulative since January 2013



A punch bowl like none before it
QT 2022/23 has barely begun



The Fed Helps To Release the Inflation Genie
US CPI and Fed Funds (July 1954 - August 2022)



Source: BLR, BLS, Bloomberg, Central Banks, NYM, Veriten. *The four central banks are BOJ, ECB, Fed, and PBOC. The decline between Feb-22 and Aug-22 reflects both maturity roll-offs and foreign exchange effects. Note: in the past 68 years, the US CPI Y/Y rate has exceeded 8% eight percent of the time. During those periods, excluding 2022, the Fed Funds Effective Rate (FFER) has ranged between a low of 5.2% and a high of 19.1%, with a mean value of 11.7% and a median value of 10.9%. As of November 4, 2022, FFER is 3.83%, following a 75-bp policy hike on 1-Nov-2022.

Global Economic Activity Is Decelerating…
…But Still Not Fast Enough Nor Far Enough to Quell Inflation

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Composite PMI SA	Nov-19	Dec-19	Jan-20	Feb-20	Mar-20	Apr-20	May-20	Jun-20	Jul-20	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-21	Feb-21	Mar-21	Apr-21
UNITED STATES	48.1	47.9	51.4	50.3	49.1	41.6	43.5	52.4	53.9	55.4	55.4	58.8	57.3	60.5	59.4	60.9	63.7	60.6
CHINA	51.8	51.5	51.1	40.3	50.1	49.4	50.7	51.2	52.8	53.1	53.0	53.6	54.9	53.0	51.5	50.9	50.6	51.9
JAPAN	48.9	48.4	48.8	47.8	44.8	41.9	38.4	40.1	45.2	47.2	47.7	48.7	49.0	50.0	49.8	51.4	52.7	53.6
GERMANY	44.1	43.7	45.3	48.0	45.4	34.5	36.6	45.2	51.0	52.2	56.4	58.2	57.8	58.3	57.1	60.7	66.6	66.2
UNITED KINGDOM	48.9	47.5	50.0	51.7	47.8	32.6	40.7	50.1	53.3	55.2	54.1	53.7	55.6	57.5	54.1	55.1	58.9	60.9
FRANCE	51.7	50.4	51.1	49.8	43.2	31.5	40.6	52.3	52.4	49.8	51.2	51.3	49.6	51.1	51.6	56.1	59.3	58.9
BRAZIL	52.9	50.2	51.0	52.3	48.4	36.0	38.3	51.6	58.2	64.7	64.9	66.7	64.0	61.5	56.5	58.4	52.8	52.3
ITALY	47.6	46.2	48.9	48.7	40.3	31.1	45.4	47.5	51.9	53.1	53.2	53.8	51.5	52.8	55.1	56.9	59.8	60.7
INDIA	51.2	52.7	55.3	54.5	51.8	27.4	30.8	47.2	46.0	52.0	56.8	58.9	56.3	56.4	57.7	57.5	55.4	55.5
RUSSIA	45.6	47.5	47.9	48.2	47.5	31.3	36.2	49.4	48.4	51.1	48.9	46.9	46.3	49.7	50.9	51.5	51.1	50.4
CANADA	51.4	50.4	50.6	51.8	46.1	33.0	40.6	47.8	52.9	55.1	56.0	55.5	55.8	57.9	54.4	54.8	58.5	57.2
AUSTRALIA	48.7	38.3	41.7	29.3	37.3	36.8	35.9	42.3	43.6	52.8	46.0	42.9	55.6	50.6	50.6	44.5	56.0	53.2
SOUTH KOREA	49.4	50.1	49.8	48.7	44.2	41.6	41.3	43.4	46.9	48.5	49.8	51.2	52.9	52.9	53.2	55.3	55.3	54.6
SPAIN	47.5	47.4	48.5	50.4	45.7	30.8	38.3	49.0	53.5	49.9	50.8	52.5	49.8	51.0	49.3	52.9	56.9	57.7
MEXICO	48.0	47.1	49.0	50.0	47.9	35.0	38.3	38.6	40.4	41.3	42.1	43.6	43.7	42.4	43.0	44.2	45.6	48.4
INDONESIA	48.2	49.5	49.3	51.9	45.3	27.5	28.6	39.1	46.9	50.8	47.2	47.8	50.6	51.3	52.2	50.9	53.2	54.6
NETHERLANDS	49.6	48.3	49.9	52.9	50.5	41.3	40.5	45.2	47.9	52.3	52.5	50.4	54.4	58.2	58.8	59.6	64.7	67.2
TURKEY	49.5	49.5	51.3	52.4	48.1	33.4	40.9	53.9	56.9	54.3	52.8	53.9	51.4	50.8	54.4	51.7	52.6	50.4
SAUDI ARABIA	58.3	56.9	54.9	52.5	42.4	44.4	48.1	47.7	50.0	48.8	50.7	51.0	54.7	57.0	57.1	53.9	53.3	55.2
SWITZERLAND	47.9	48.4	48.0	49.2	43.5	41.2	42.5	41.4	49.6	50.6	52.5	53.1	55.4	58.6	60.1	62.0	65.6	68.3
SWEDEN	46.0	46.6	51.0	52.5	43.5	36.0	39.9	49.1	52.1	55.4	56.6	58.8	59.6	64.4	62.0	61.8	64.3	68.7
POLAND	46.7	48.0	47.4	48.2	42.4	31.9	40.6	47.2	52.8	50.6	50.8	50.8	50.8	51.7	51.9	53.4	54.3	53.7
TAIWAN	49.8	50.8	51.8	49.9	50.4	42.2	41.9	46.2	50.6	52.2	55.2	55.1	56.9	59.4	60.2	60.4	60.8	62.4
NORWAY	54.0	54.4	51.5	52.9	42.3	42.9	46.6	50.1	45.3	47.9	51.1	54.1	52.7	53.0	52.4	57.4	60.8	59.1
AUSTRIA	46.0	46.0	49.2	50.2	45.8	31.6	40.4	46.5	52.8	51.0	51.7	54.0	51.7	53.5	54.2	58.3	63.4	64.7
UAE	50.3	50.2	49.3	49.1	45.2	44.1	46.7	50.4	50.8	49.4	51.0	49.5	49.5	51.2	51.2	50.6	52.6	52.7
SOUTH AFRICA	48.6	47.6	48.3	48.4	44.5	35.1	32.5	42.5	44.9	45.3	49.4	51.0	50.3	50.2	50.8	50.2	50.3	53.7
MALAYSIA	49.5	50.0	48.8	48.5	48.4	31.3	45.6	51.0	50.0	49.3	49.0	48.5	48.4	49.1	48.9	47.7	49.9	53.9
SINGAPORE	50.4	51.0	51.4	47.0	33.3	28.1	27.1	43.2	45.6	43.6	45.1	48.6	46.7	50.5	52.9	54.9	53.5	51.8
ISRAEL	52.8	51.8	50.1	0.0	34.6	39.3	38.5	45.0	49.8	53.1	53.5	54.9	57.1	50.9	49.5	49.4	48.6	52.6
EGYPT	47.9	48.2	46.0	47.1	44.2	29.7	40.7	44.6	49.6	49.4	50.4	51.4	50.9	48.2	48.7	49.3	48.0	47.7
IRELAND	49.7	49.5	51.4	51.2	45.1	36.0	39.2	51.0	57.3	52.3	50.0	50.3	52.2	57.2	51.8	52.0	57.1	60.8
GREECE	54.1	53.9	54.4	56.2	42.5	29.5	41.1	49.4	48.6	49.4	50.0	48.7	42.3	46.9	50.0	49.4	51.8	54.4
CZECH REPUBLIC	43.5	43.6	45.2	46.5	41.3	35.1	39.6	44.9	47.0	49.1	50.7	51.9	53.9	57.0	57.0	56.5	58.0	58.9
VIETNAM	51.0	50.8	50.6	49.0	41.9	32.7	42.7	51.1	47.6	45.7	52.2	51.8	49.9	51.7	51.3	51.6	53.6	54.7
HUNGARY	53.1	53.9	52.2	50.3	28.5	33.7	40.7	47.4	51.1	52.3	48.7	50.2	51.9	51.6	54.5	49.0	48.8	51.0
KENYA	53.2	53.3	49.7	49.0	37.5	34.8	36.7	46.6	54.2	53.0	56.3	59.1	51.3	51.4	53.2	50.9	50.6	41.5
HONG KONG	38.5	42.1	46.8	33.1	34.9	36.9	43.9	49.6	44.5	44.0	47.7	49.8	50.1	43.5	47.8	50.2	50.5	50.3
DEVELOPED MARKETS	49.5	49.1	49.8	49.5	45.9	36.8	39.5	46.4	49.8	51.2	52.2	52.8	53.8	54.8	55.2	56.5	58.5	59.3
EMERGING MARKETS	51.0	51.0	51.0	44.6	49.1	42.7	45.4	49.6	51.4	52.5	52.8	53.4	53.9	52.8	52.1	51.6	51.3	52.2
GLOBAL PMI	50.3	50.1	50.4	47.1	47.3	39.6	42.4	48.0	50.6	51.8	52.4	53.1	53.8	53.8	53.6	54.0	54.9	55.8

Composite PMI SA	May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21	Jan-22	Feb-22	Mar-22	Apr-22	May-22	Jun-22	Jul-22	Aug-22	Sep-22	Oct-22
UNITED STATES	61.6	60.9	59.9	59.7	60.5	60.8	60.6	58.8	57.6	58.6	57.1	55.4	56.1	53.0	52.8	52.8	50.9	50.2
CHINA	52.0	51.3	50.3	49.2	50.0	50.6	49.9	50.9	49.1	50.4	48.1	46.0	48.1	51.7	50.4	49.5	48.1	49.2
JAPAN	53.0	52.4	53.0	52.7	51.5	53.2	54.5	54.3	55.4	52.7	54.1	53.5	53.3	52.7	52.1	51.5	50.8	50.7
GERMANY	64.4	65.1	65.9	62.6	58.4	57.8	57.4	57.4	59.8	58.4	56.9	54.6	54.8	52.0	49.3	49.1	47.8	45.1
UNITED KINGDOM	65.6	63.9	60.4	60.3	57.1	57.8	58.1	57.9	57.3	58.0	55.2	55.8	54.6	52.8	52.1	47.3	48.4	46.2
FRANCE	59.4	59.0	58.0	57.5	55.0	53.6	55.9	55.6	55.5	57.2	54.7	55.7	54.6	51.4	49.5	50.6	47.7	47.2
BRAZIL	53.7	56.4	56.7	53.6	54.4	51.7	49.8	49.8	47.8	49.6	52.3	51.8	54.2	54.1	54.0	51.9	51.1	50.8
ITALY	62.3	62.2	60.3	60.9	59.7	61.1	62.8	62.0	58.3	58.3	55.8	54.5	51.9	50.9	48.5	48.0	48.3	46.5
INDIA	50.8	48.1	55.3	52.3	53.7	55.9	57.6	55.5	54.0	54.9	54.0	54.7	54.6	53.9	56.4	56.2	55.1	55.3
RUSSIA	51.9	49.2	47.5	46.5	49.8	51.6	51.7	51.6	51.8	48.6	44.1	48.2	50.8	50.9	50.3	51.7	52.0	50.7
CANADA	57.0	56.5	56.2	57.2	57.0	57.7	57.2	56.5	56.2	56.6	58.9	56.2	56.8	54.6	52.5	48.7	49.8	48.8
AUSTRALIA	57.6	63.4	61.8	45.5	54.2	48.7	56.4	48.8	48.8	51.6	50.7	53.9	56.9	40.7	51.1	42.4	44.4	42.3
SOUTH KOREA	53.7	53.9	53.0	51.2	52.4	50.2	50.9	51.9	52.8	53.8	51.2	52.1	51.8	51.3	49.8	47.6	47.3	48.2
SPAIN	59.4	60.4	59.0	59.5	58.1	57.4	57.1	56.2	56.2	56.9	54.2	53.3	53.8	52.6	48.7	49.9	49.0	44.7
MEXICO	47.6	48.8	49.6	47.1	48.6	49.3	49.4	49.4	46.1	48.0	49.2	49.3	50.6	52.2	48.5	48.5	50.3	50.3
INDONESIA	55.3	53.5	40.1	43.7	52.2	57.2	53.9	53.5	53.7	51.2	51.3	51.9	50.8	50.2	51.3	51.7	53.7	51.8
NETHERLANDS	69.4	68.8	67.4	65.8	62.0	62.5	60.7	58.7	60.1	60.6	58.4	59.9	57.8	55.9	54.5	52.6	49.0	47.9
TURKEY	49.3	51.3	54.0	54.1	52.5	51.2	52.0	52.1	50.5	50.4	49.4	49.2	49.2	48.1	46.9	47.4	46.9	46.4
SAUDI ARABIA	56.4	56.4	55.8	54.1	58.6	57.7	56.9	53.9	53.2	56.2	56.8	55.7	55.7	57.0	56.3	57.7	56.6	57.2
SWITZERLAND	68.6	67.1	70.0	67.0	67.6	65.7	63.8	64.2	63.8	62.6	64.0	62.5	60.0	59.1	58.0	56.4	57.1	54.9
SWEDEN	65.8	65.8	65.0	60.7	64.6	64.3	62.8	61.2	61.8	57.8	57.0	54.6	54.5	53.1	52.2	49.8	48.9	46.8
POLAND	57.2	59.4	57.6	56.0	53.4	53.8	54.4	56.1	54.5	54.7	52.7	52.4	48.5	44.4	42.1	40.9	43.0	42.0
TAIWAN	62.0	57.6	59.7	58.5	54.7	55.2	54.9	55.5	55.1	54.3	54.1	51.7	50.0	49.8	44.6	42.7	42.2	41.5
NORWAY	58.4	60.8	62.5	61.3	58.8	58.5	63.6	57.7	56.7	55.8	58.9	59.4	54.3	55.4	53.6	51.9	50.3	53.1
AUSTRIA	66.4	67.0	63.9	61.8	62.8	60.6	58.1	58.7	61.5	58.4	59.3	57.9	56.6	51.2	51.7	48.8	48.8	46.6
UAE	52.3	52.2	54.0	53.8	53.3	55.7	55.9	55.6	54.1	54.8	54.8	54.6	55.6	54.8	55.4	56.7	56.1	56.6
SOUTH AFRICA	53.2	51.0	46.1	49.9	50.7	48.6	51.7	48.4	50.9	50.9	51.4	50.3	50.7	52.5	52.7	51.7	49.2	49.5
MALAYSIA	51.3	39.9	40.1	43.4	48.1	52.2	52.3	52.8	50.5	50.9	49.6	51.6	50.1	50.4	50.6	50.3	49.1	48.7
SINGAPORE	54.4	50.1	56.7	52.1	53.8	52.3	52.0	55.1	54.4	52.5	52.9	56.7	59.4	57.5	58.0	56.0	57.5	57.7
ISRAEL	56.1	51.5	46.8	54.1	58.1	55.4	50.3	47.8	42.9	49.7	47.4	51.2	52.6	53.4	57.6	49.5	51.3	
EGYPT	48.6	49.9	49.1	49.8	48.9	48.7	48.7	49.0	47.9	48.1	46.5	46.9	47.0	45.2	46.4	47.6	47.6	47.7
IRELAND	64.1	64.0	63.3	62.8	60.3	62.1	59.9	58.3	59.4	57.8	59.4	59.1	56.4	53.1	51.8	51.1	51.5	51.4
GREECE	58.0	58.6	57.4	59.3	58.4	58.9	58.8	59.0	57.9	57.8	54.6	54.8	53.8	51.1	49.1	48.8	49.7	48.1
CZECH REPUBLIC	61.8	62.7	62.0	61.0	58.0	55.1	57.1	59.1	59.0	56.5	54.7	54.4	52.3	49.0	46.8	46.8	44.7	41.7
VIETNAM	53.1	44.1	45.1	40.2	40.2	52.1	52.2	52.5	53.7	54.3	51.7	51.7	54.7	54.0	51.2	52.7	52.5	50.6
HUNGARY	53.1	55.0	55.8	55.6	52.0	52.9	52.2	65.5	50.9	53.8	57.6	57.6	51.9	57.0	58.0	57.8	49.6	56.4
KENYA	52.5	51.0	50.6	51.1	50.4	51.4	53.0	53.7	47.6	52.9	50.5	49.5	48.2	46.8	46.3	44.2	51.7	50.2
HONG KONG	52.5	51.4	51.3	53.3	51.7	50.8	52.6	50.8	48.9	42.9	42.0	51.7	54.9	52.4	52.3	51.2	48.0	49.3
DEVELOPED MARKETS	59.8	59.5	59.8	58.3	57.1	56.6	56.9	56.6	56.3	56.6	56.5	56.3	55.0	52.5	51.3	50.3	50.1	48.8
EMERGING MARKETS	52.0	51.3	50.6	49.6	50.8	51.6	51.2	51.7	50.0	50.9	49.2	48.1	49.5	51.7	50.8	50.2	49.4	49.8
GLOBAL PMI	56.0	55.5	55.4	54.1	54.1	54.2	54.2	54.3	53.2	53.7	52.9	52.3	52.3	52.2	51.1	50.3	49.8	49.4

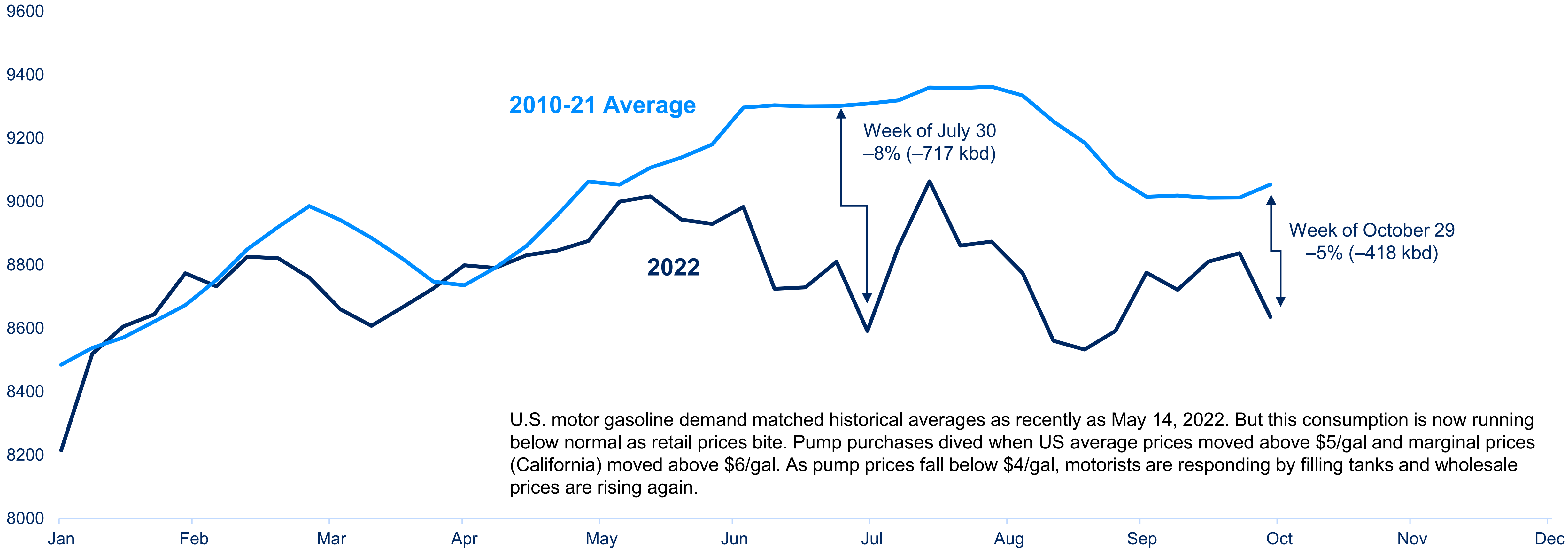
Source: JPM, IHS-Markit, Credit Suisse, Swedbank Markets, DNB, HALPIM, BLR. Note: as of 4-Nov-2022, Oct-22 figures for Israel are not yet available.

U.S. Gasoline Demand Is Already Exhibiting A Meaningful Slump

VERITEN

U.S. Gasoline Demand

thousand b/d (smoothed by four-week moving averages)

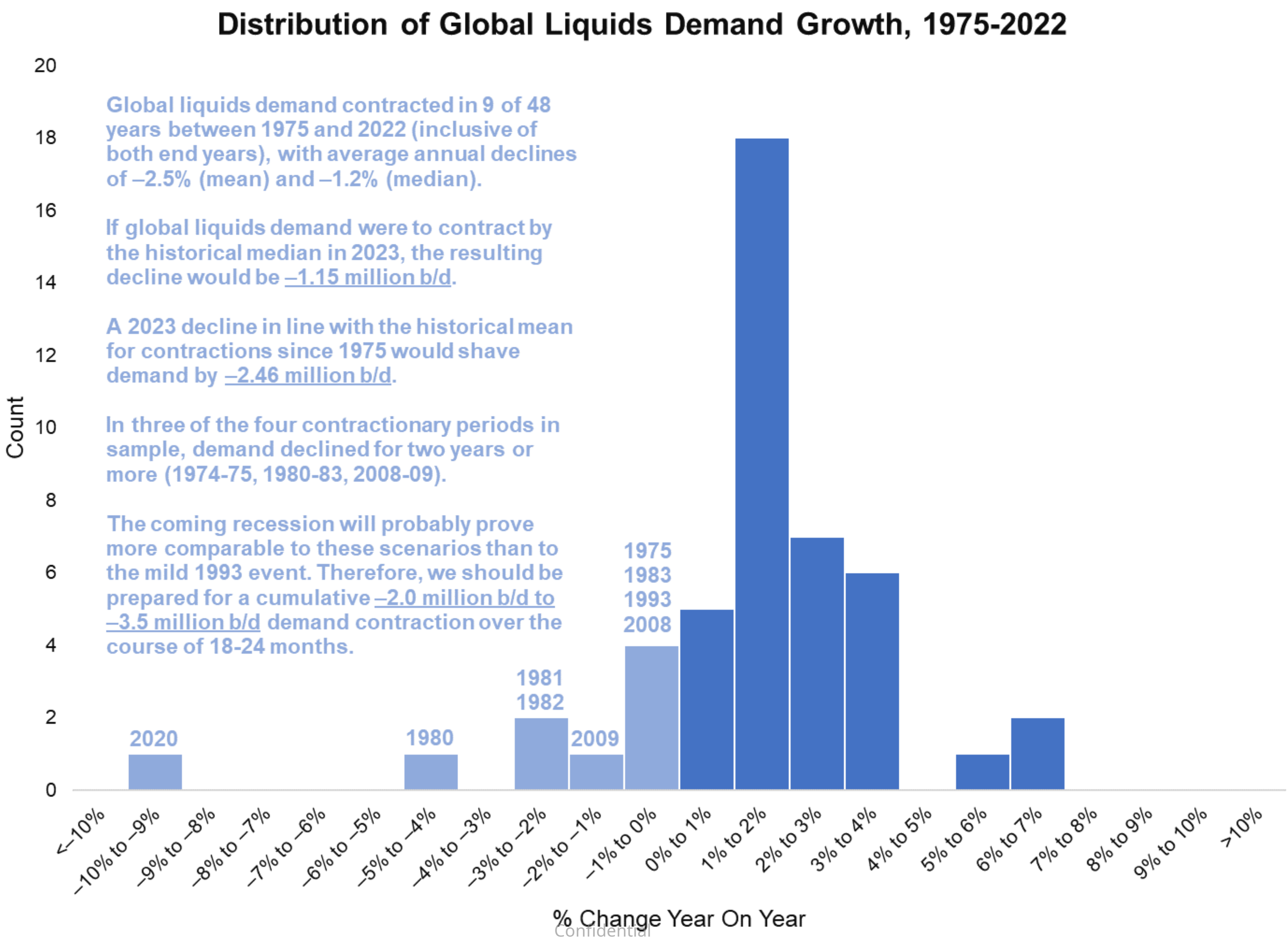


U.S. motor gasoline demand matched historical averages as recently as May 14, 2022. But this consumption is now running below normal as retail prices bite. Pump purchases dived when US average prices moved above $5/gal and marginal prices (California) moved above $6/gal. As pump prices fall below $4/gal, motorists are responding by filling tanks and wholesale prices are rising again.

Source: EIA, BLR, Veriten.

How Bad Would A Recession Be For Oil Demand?

VERITEN

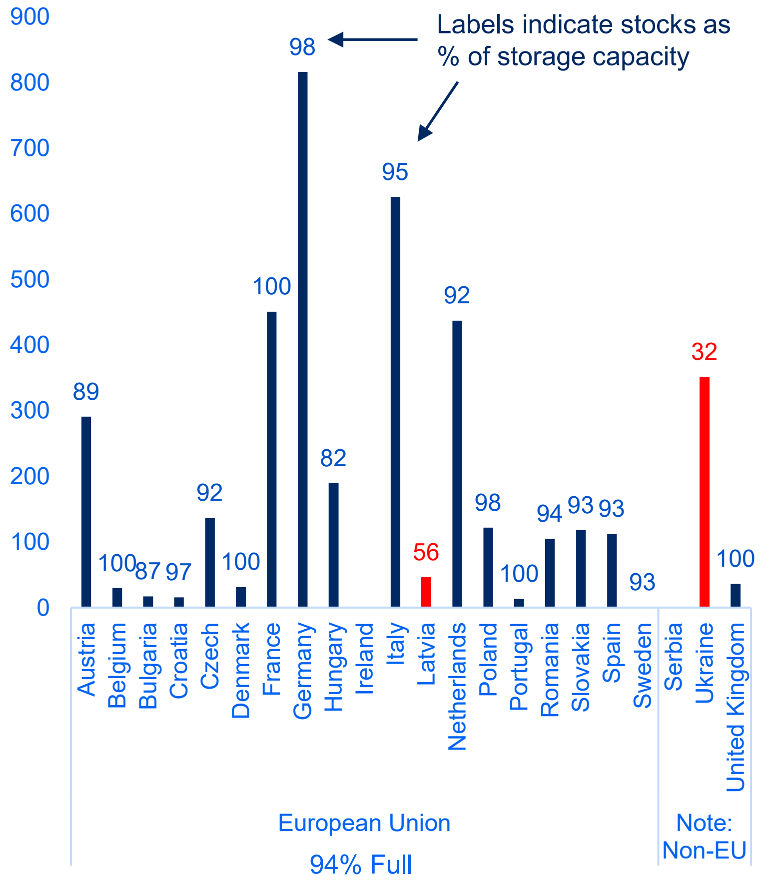


Source: BLR, Veriten.

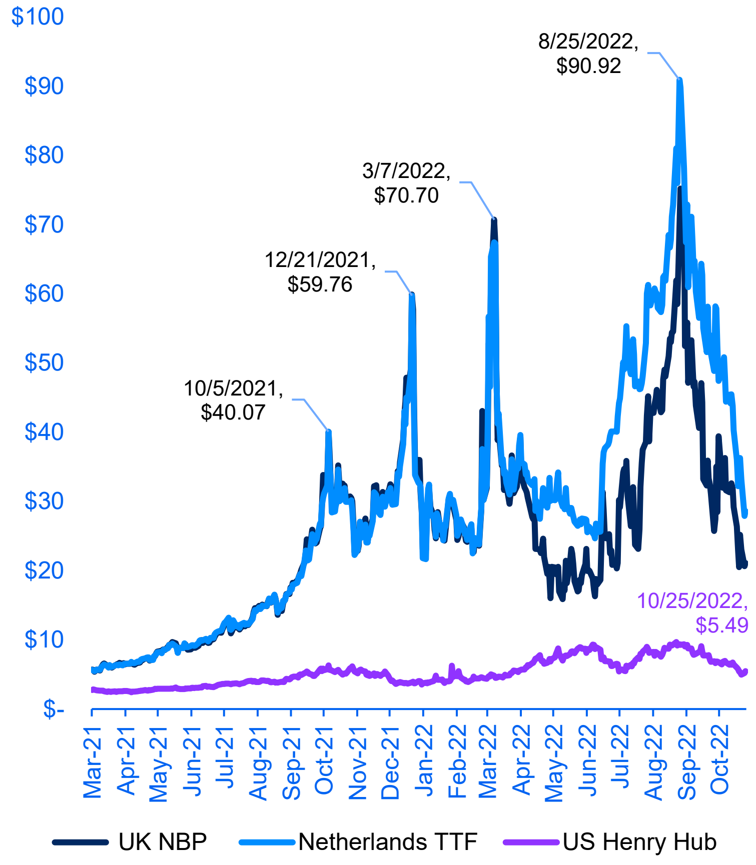
EU Gas Developments Force Subzero Gas Prices in West Texas, But…

VERITEN

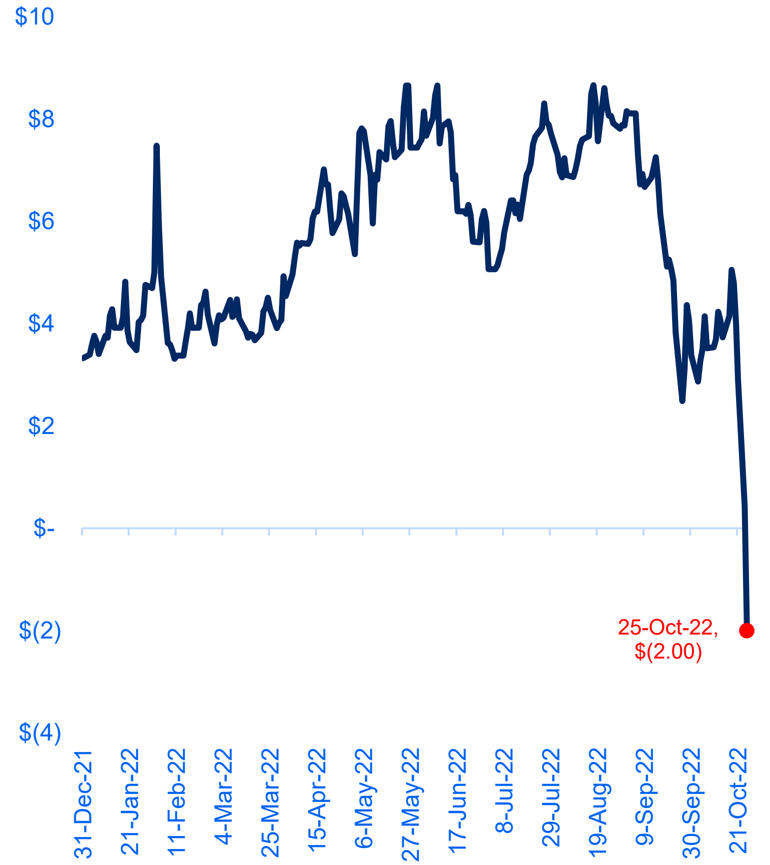
Natural Gas in Storage in Europe
Bcf by country (as of 23 October 2022)



Natural Gas Marker Prices
US$ per MMBtu (as of 25 October 2022)



Waha Hub, Spot Natural Gas Price
$ per MMBtu (as of 25 October 2022)

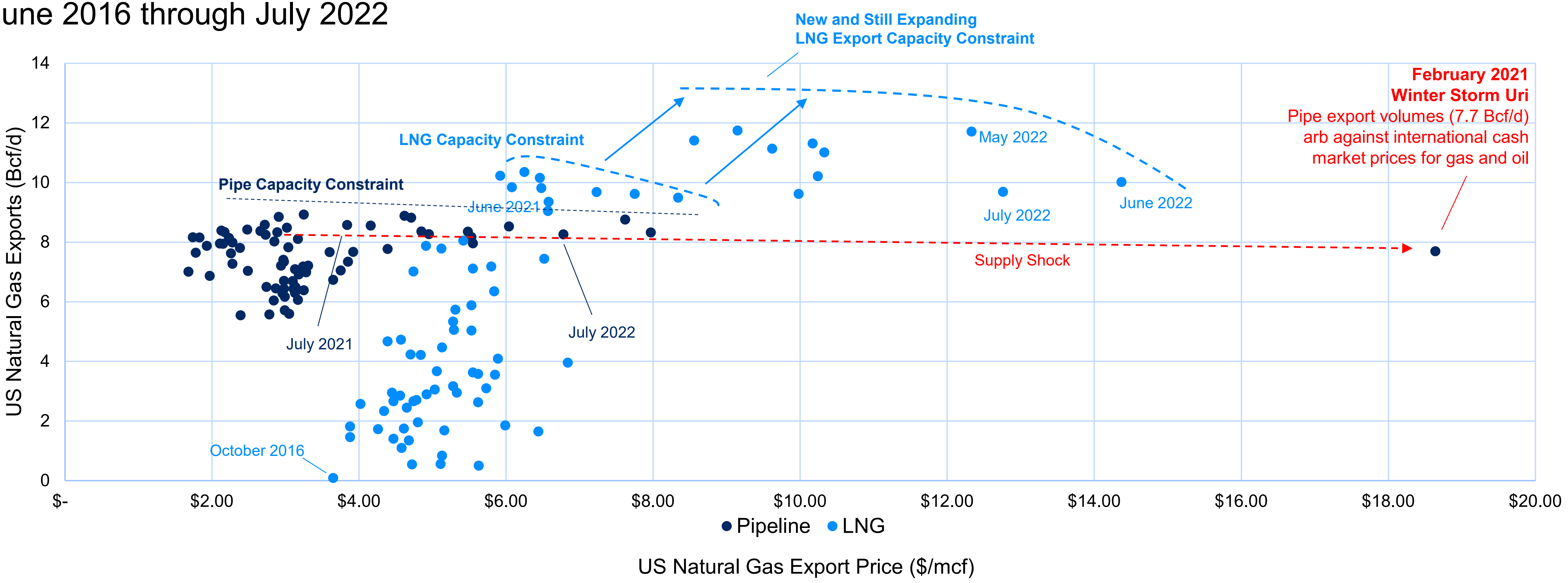


Source: GIE, ALSI, ICE, Bloomberg, Veriten. Note: storage relative to storage capacity can provide a false sense of security. Measured against normal annual gas consumption, "100% full storage" covers 145 days of demand in Denmark, 112 days in France, 24 days in Portugal, 19 days in Belgium, and just 5 days in the United Kingdom. Moreover, because demand is seasonally higher in winter than summer, even these figures overstate days of immediate coverage heading into winter 2022/23. LNG stocks either in EU terminals or awaiting discharge from tankers are small in comparison.

VERITEN

U.S. Natural Gas Exports by Supply Channel

June 2016 through July 2022



Source: EIA, BLR, Veriten.

Between 2001 and 2020, EU cut gas output by 64% but lifted demand by 2%

VERITEN

EU Gas Policies Increased EU Gas Import Dependency

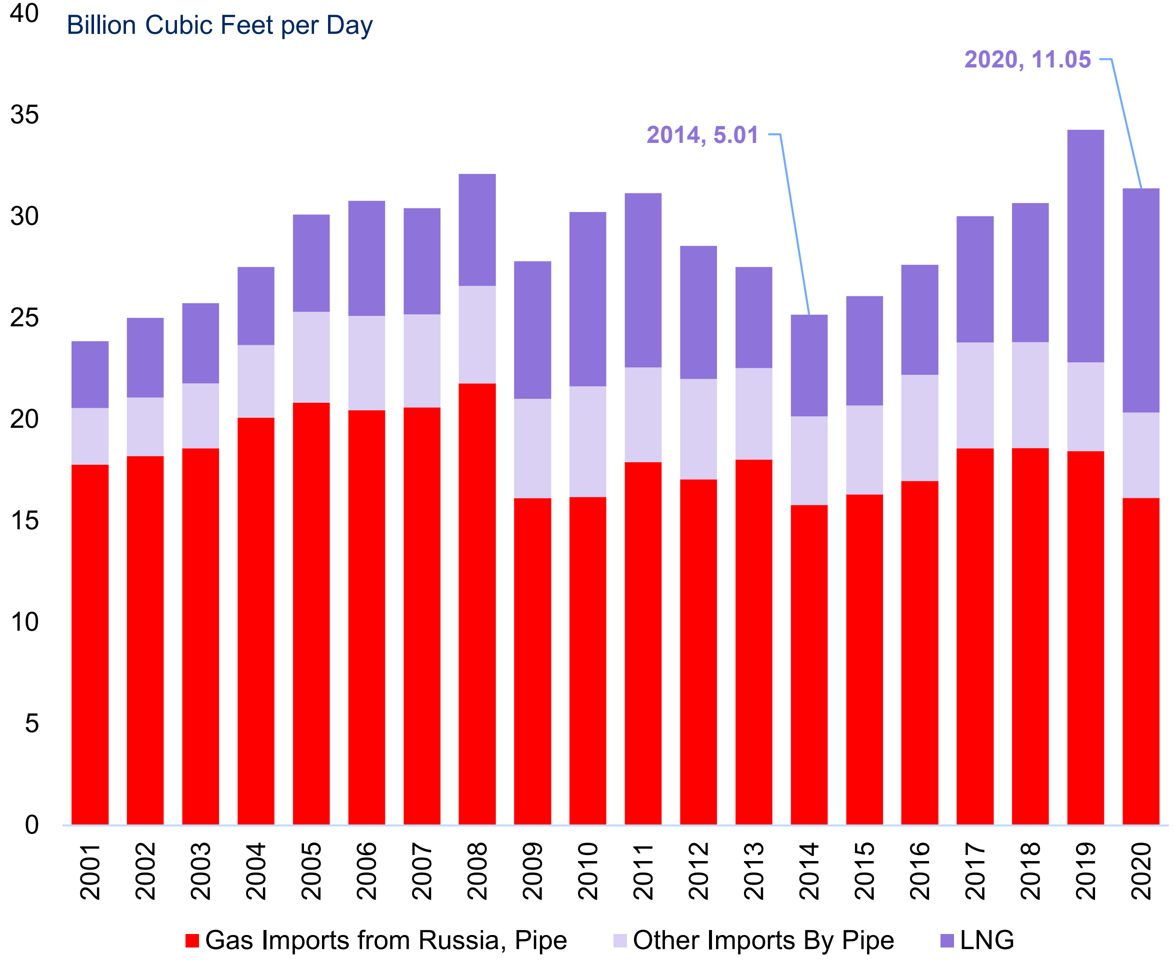
Natural Gas Balance, European Union (Bcf/d)

	2001	2014	2020	Change 2014 v 2020
Production	12.82	9.67	4.61	–52%
Consumption	35.93	32.06	36.66	+14%
Balance	–23.11	–22.39	–32.05	–9.66
Import Dependency	64%	70%	87%	+17%-pts

Natural Gas Balance, Rest of Europe (Bcf/d)

	2001	2014	2020	Change 2014 v 2020
Production	17.55	16.08	16.48	+2%
Consumption	19.17	16.32	15.55	–5%
Balance	–1.62	–0.24	+0.93	+1.17
Import Dependency	8%	1%	6% Surplus	–7%-pts

Europe Fills Its Gap With LNG & Russian Pipe Gas



*LNG from all sources, not just Russia

Source: BPSR, BLR. Note: in 2020, 85% of Russian gas exports by pipeline went to Europe (16.1 Bcf/d of 19.0 Bcf/d total).Russia shipped 3.9 Bcf/d in LNG cargoes in 2020, of which 1.7 Bcf/d (43%) went to Europe. The European Union accounted for almost all this inflow, led by France (0.48 Bcf/d), Spain (0.33 Bcf/d), and the United Kingdom (0.28 Bcf/d). Russia supplied 15% of all Europe's LNG imports in 2020.

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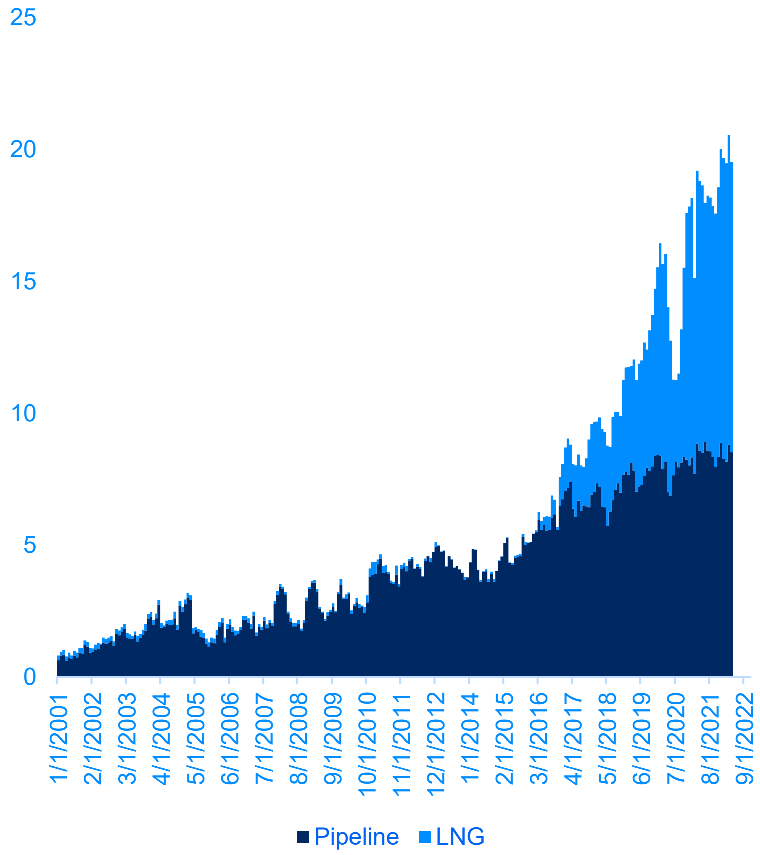
Sam Sam
Jun 26, 2024 12:06 PM EDT

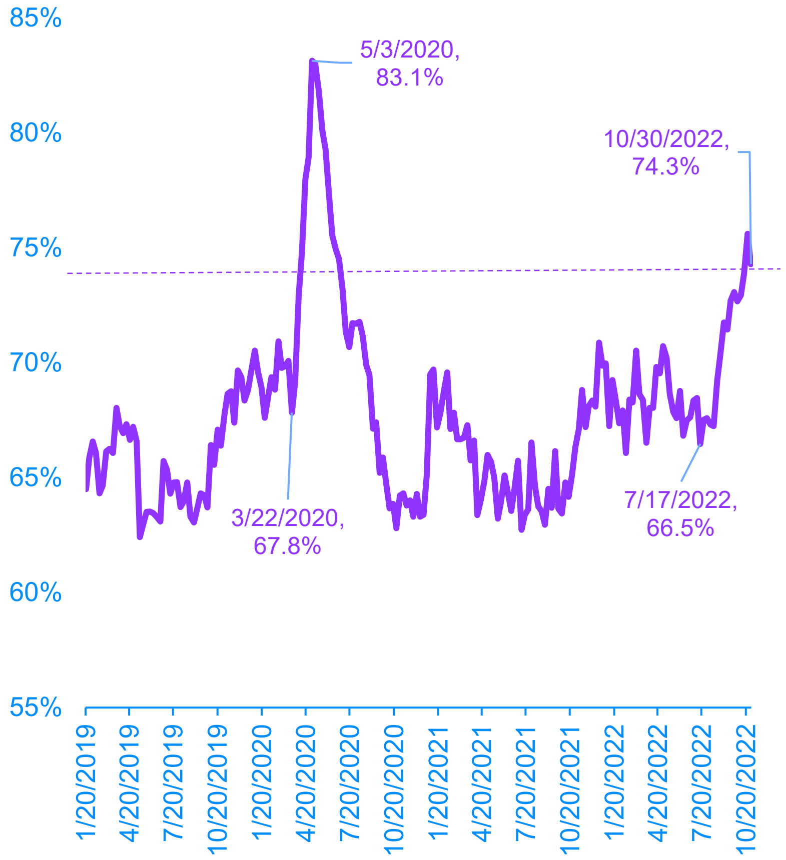
Softer End-Use Consumption, But Stronger Demand To Move Inventories

VERITEN

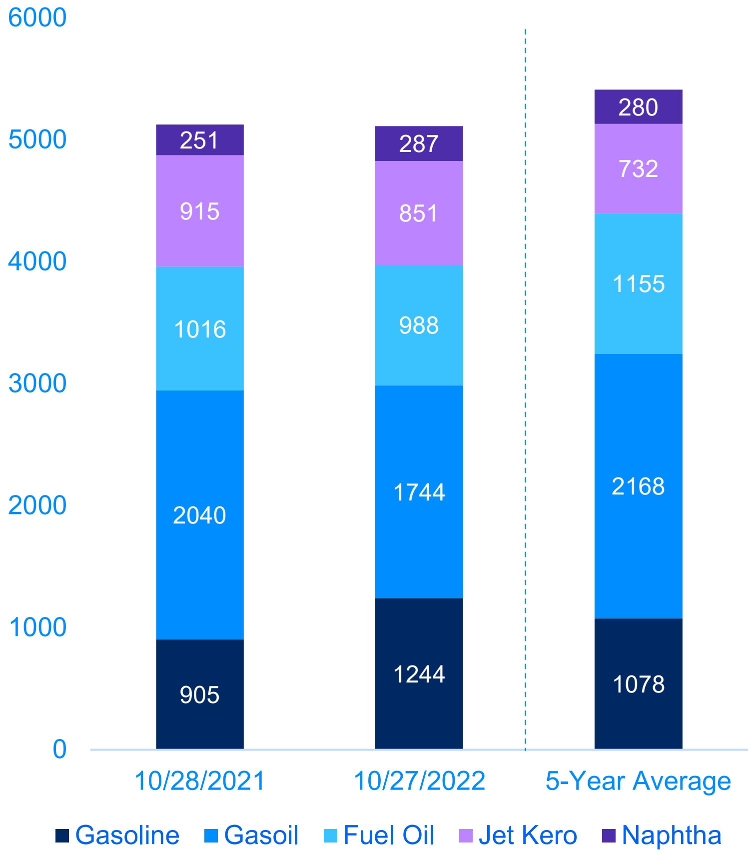
U.S. Natural Gas Exports
Bcf/d



Global VLCC Fleet is Moving
utilization of capacity



Oil Product Inventory, ARA
thousand metric tonnes



ARA = Amsterdam, Rotterdam, Antwerp

Source: EIA, Vortexa, PJK.

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Sam Sam
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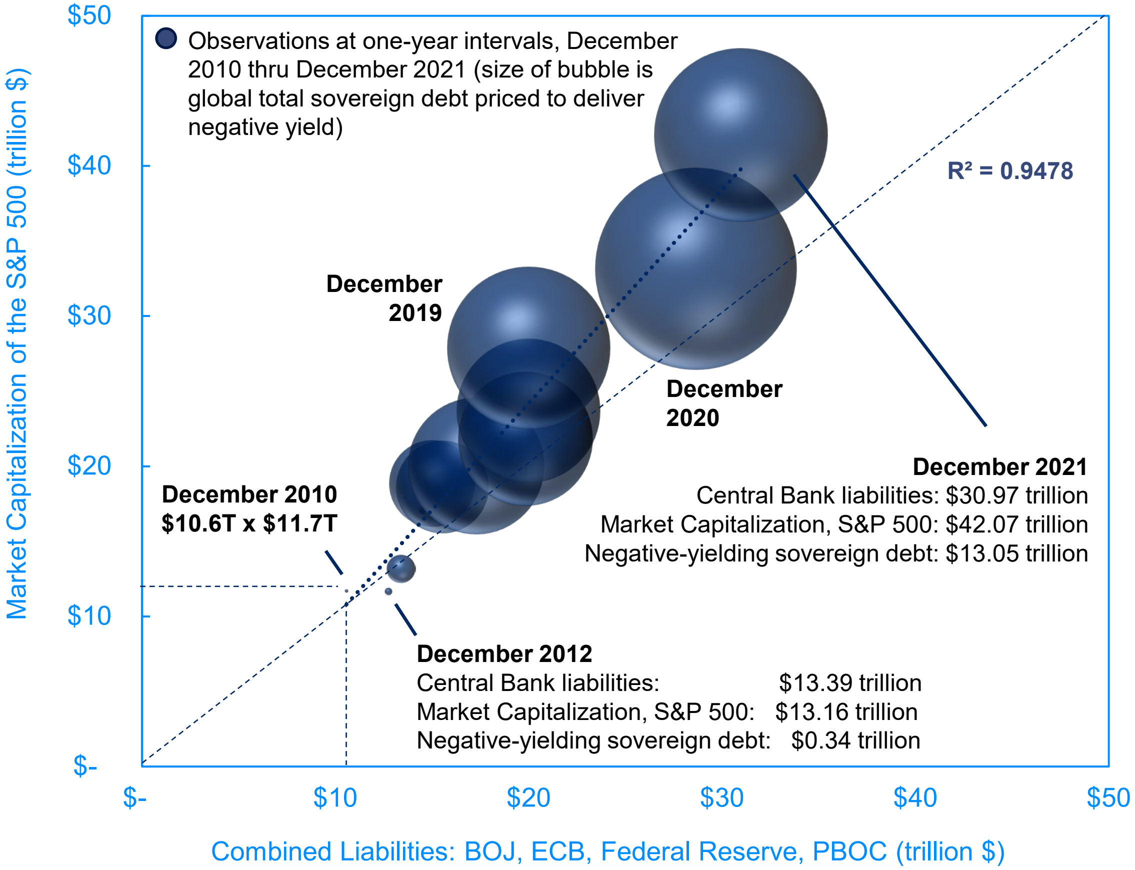
"Breaking Something" In High P/E Tech Is Not The Risk—It Is The Objective

VERITEN

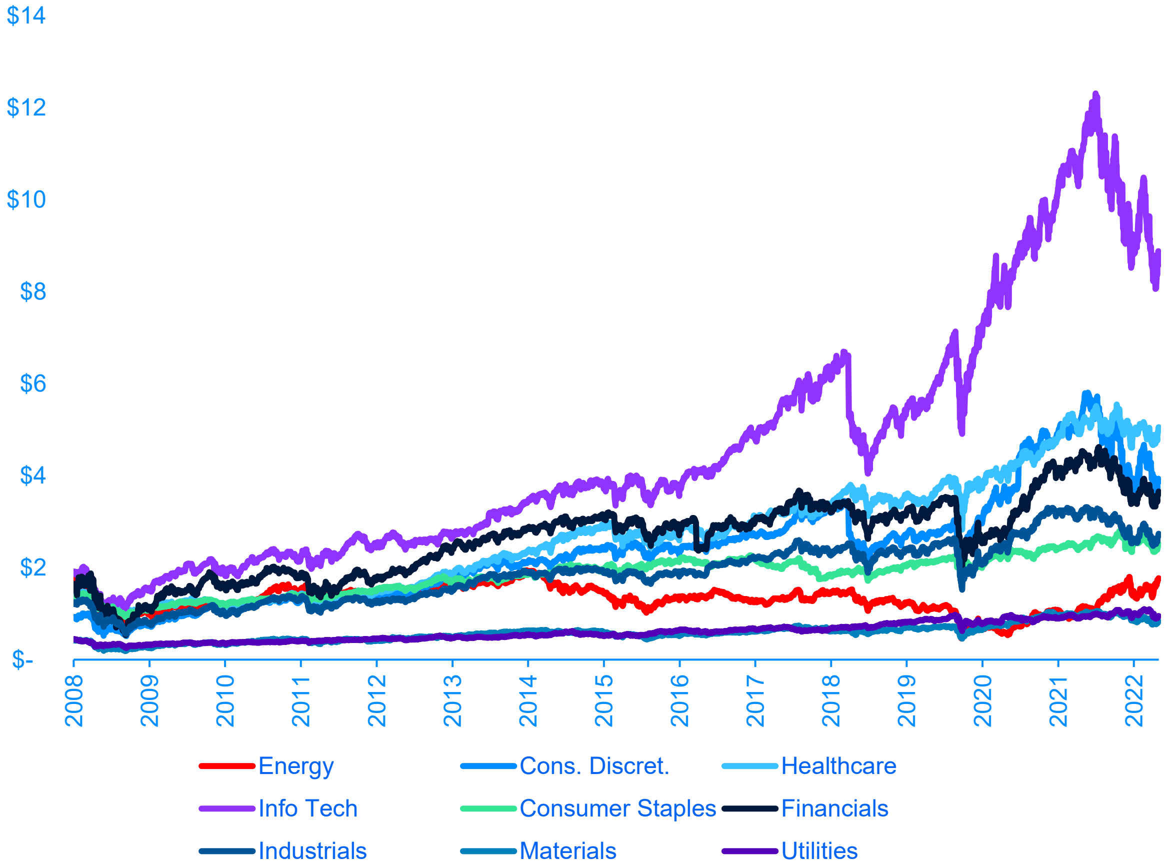
Bubble Trouble

Experimental monetary and fiscal policy inflated asset bubbles



S&P 500 Market Capitalization by Sector

trillion USD, 30-Jun-2008 through 4-Nov-2022



Source: BOJ, ECB, Federal Reserve, PBOC, Bloomberg, BLR. Note: since 1-Jan-2000, max market cap for S&P 500 Energy is $1.946 trillion; current market cap is $1.773 trillion (–8.9%). Max market cap for Info Tech in same time interval is $12.314 trillion; current market cap is $8.194 trillion (–33.5%). To reach –50% would require another $2.04 trillion in IT market cap to vanish. Energy is now 5.5% of S&P 500 market capitalization, having regained the 5% threshold on October 7, 2022 and the 3.0% threshold on January 13, 2022. Energy's weight was below 2.0% as recently as November 6, 2020.

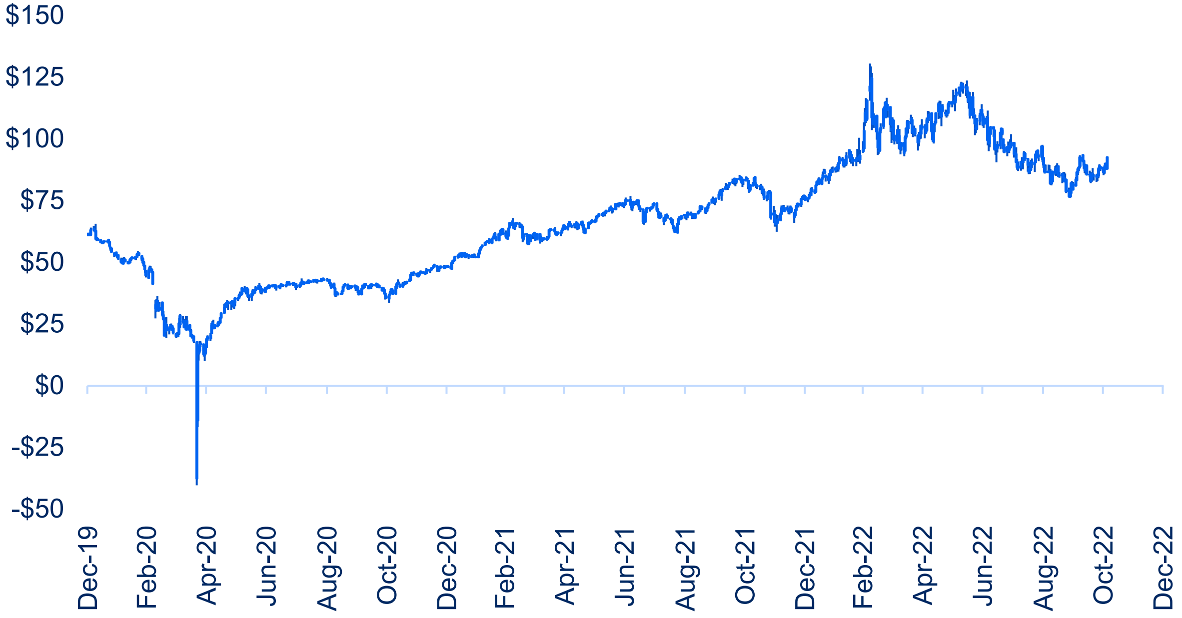
Short-Run Price Probabilities Implied By Listed Options Markets

VERITEN

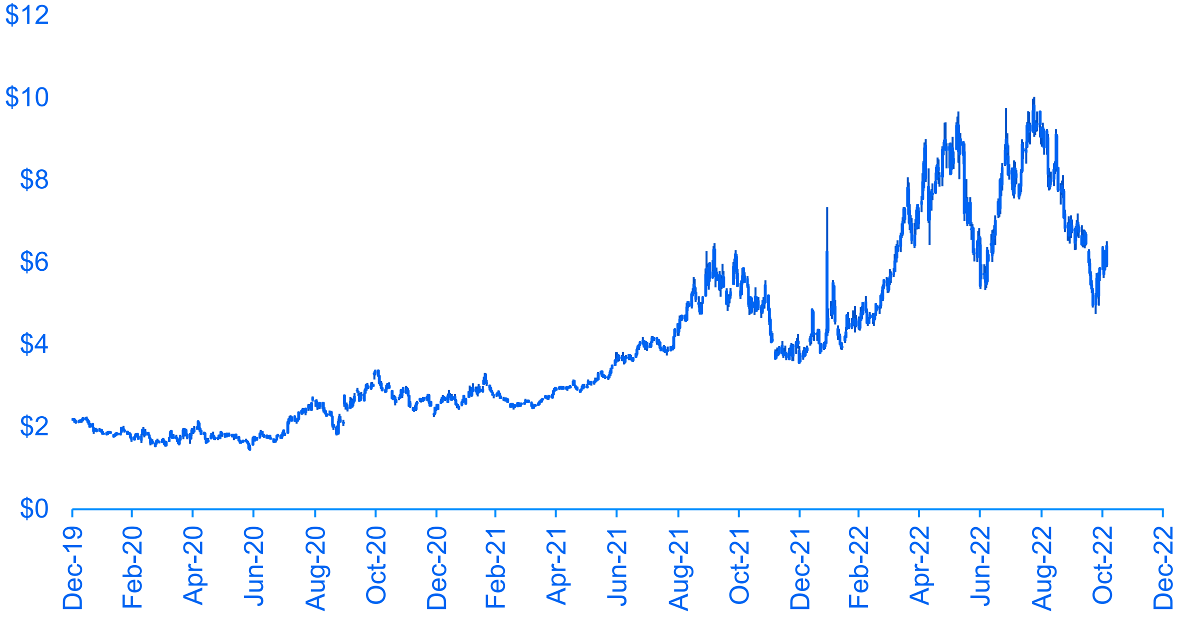
NYM WTI	< $45	$65 to $85	>$100
Dec-2022	0%	25%	24%
Jun-2023	20%	22%	21%
Dec-2023	34%	16%	20%

NYM Gas	< $4.50	$5 to $8	>$9.00
Dec-2022	14%	61%	8%
Jun-2023	57%	24%	8%
Dec-2023	46%	27%	15%

NYM WTI Crude Oil – Rolling Prompt Price
$ per bbl



NYM Natural Gas – Rolling Prompt Price
$ per MMBtu



Source: NYM, BLR. Note: as of closing prices on 4-Nov-2022.

New Biden Administration Policy Aims to Set an "Oil Price Floor"

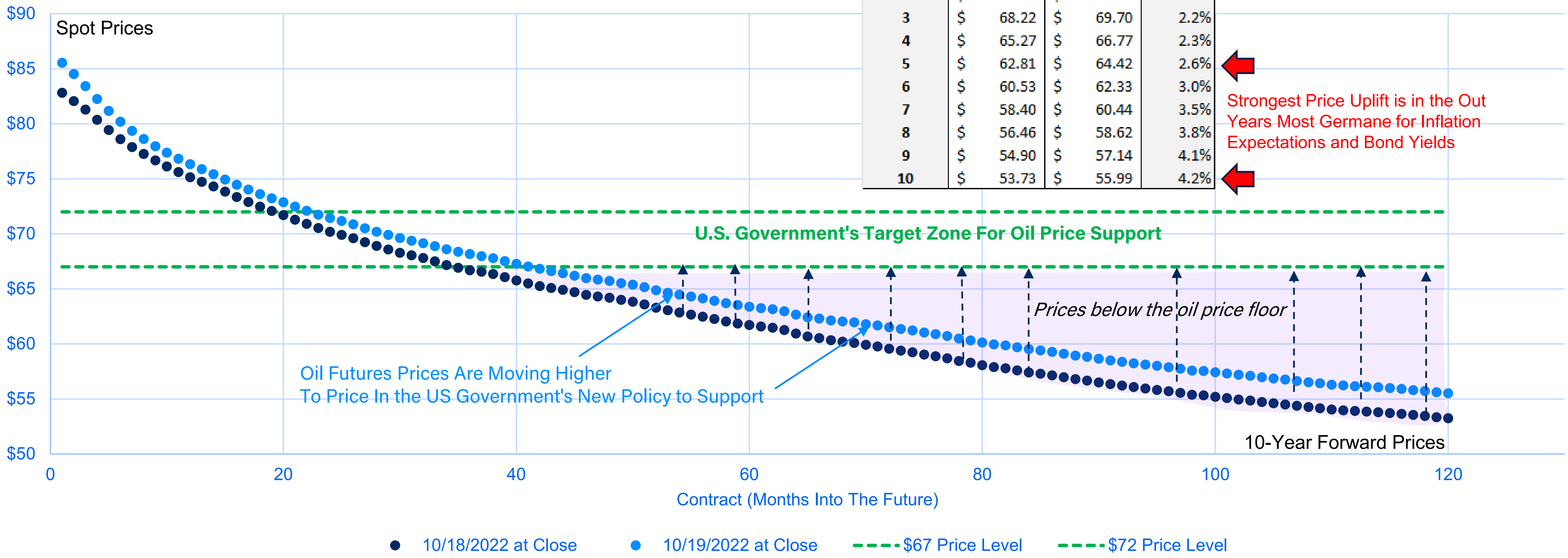
VERITEN

NYM WTI Average Price By Forward Year			
Years Forward	18-Oct-2022 Close	19-Oct-2022 Close	DoD Change
1	$ 78.61	$ 80.30	2.2%
2	$ 72.36	$ 73.52	1.6%
3	$ 68.22	$ 69.70	2.2%
4	$ 65.27	$ 66.77	2.3%
5	$ 62.81	$ 64.42	2.6%
6	$ 60.53	$ 62.33	3.0%
7	$ 58.40	$ 60.44	3.5%
8	$ 56.46	$ 58.62	3.8%
9	$ 54.90	$ 57.14	4.1%
10	$ 53.73	$ 55.99	4.2%

Strongest Price Uplift is in the Out Years Most Germane for Inflation Expectations and Bond Yields

NYM WTI Crude Oil Futures Curve

$ per barrel



Source: NYM, BLR.

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Conclusions

- Events Continue to Validate the ‘Folly and Hammer’ Investment Thesis We Presented At EnCap’s Annual LP Meeting in October 2018
- Private Investments in Oil and Gas Are Likely to Deliver Superior Absolute and Relative Returns Over the Next 5+ Years

Hydrocarbons Boom in the Year of Folly and the Hammer |

Colin P. Fenton
Blacklight Research, LLC
October 22, 2018 | Dallas

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